# Childhood Leukemia

## About childhood leukemia

**Leukemias** are cancers of the blood cells. Leukemia most often occurs in the earliest forms of white blood cells.

The center of our bones, called the bone marrow[1], makes our red blood cells, white blood cells, and platelets. When leukemia occurs, cancer cells multiply in the bone marrow. As the cancer cells build up, they crowd out the normal cells. When this happens, leukemia cells can spill into the bloodstream and the bone marrow can have a tough time making enough normal cells. This can cause people with leukemia to have symptoms like low energy, bone pain, infections, easy bleeding, or bruising.

Sometimes leukemia can spread to other parts of the body such as the lymph nodes, spleen, liver, central nervous system (the brain and spinal cord), testicles, or other organs.

Some other childhood cancers, such as neuroblastoma[2] or rhabdomyosarcoma[3], start in other organs and can spread to the bone marrow, but these cancers are not leukemia.

### Key Statistics for Childhood Leukemia

The American Cancer Society's most recent estimates and statistics for childhood leukemia in the United States are located here.

### Causes, Risk Factors, and Prevention of Childhood Leukemia

Learn about the risk factors for childhood leukemia and if there are things that can be done to help lower the risk.

Early Detection, Diagnosis, and Prognosis of Childhood Leukemia
Learn about the signs and symptoms of childhood leukemia, the tests that may be done for it, and the outlook for those with a diagnosis.

Treating Childhood Leukemia
If your child is facing childhood leukemia, we can help you learn about the treatment options and possible side effects and point you to information and services to help.

After Treatment for Childhood Leukemia
Get information about how to live well after childhood leukemia treatment and make decisions about next steps.

## Types of leukemia in children

There are a few types of leukemia that affect children. The type of leukemia a child has plays a key role in how it is treated and the child's outlook (prognosis). To diagnose leukemia, testing on samples of the blood, bone marrow, cerebrospinal fluid (CSF), and sometimes lymph nodes may be needed.

Be sure to ask the health care team if you have any questions about your child's leukemia diagnosis.

The different types of leukemia are based on:

- If the leukemia is **acute** (fast growing) or **chronic** (slower growing)
- If the leukemia starts in **lymphoid** cells or **myeloid** cells, both of which are types of immune cells that are made in the bone marrow

### Acute leukemias

Most childhood leukemias are acute. These leukemias grow quickly and often need treatment right away. The main types of acute leukemia are acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML). In rare cases, acute leukemia can have features of both ALL and AML. When this happens, it is called a mixed phenotype acute leukemia (MPAL).

## Acute lymphoblastic (lymphocytic) leukemia (ALL)

About 3 out of 4 childhood leukemias are ALL. These leukemias start in early forms of white blood cells, called **lymphoblasts**. Normally they would mature into certain cells that help our bodies fight infections, called **lymphocytes**. There are several subtypes of ALL, which are based mainly on:

- The type of cell the leukemia looks most like (often B cell or T cell) and how far the cells matured before becoming cancer, known as the **immunophenotype** of the leukemia
- If the cancer cells have certain **gene or chromosome changes**

### *B-cell ALL*

The most common form of ALL in children starts in early forms of B cells, which normally mature into cells that make antibodies to fight infections. When these cells develop gene changes or an abnormal number of chromosomes, they can become cancer. These B cells lose the ability to fight infections and can multiply very quickly.

B cells travel to the spleen (an immune organ in the belly, usually under the left ribcage) and lymph nodes to grow and mature. When B cells become leukemia, the spleen and lymph nodes may enlarge. This can result in a large spleen or large lymph nodes, commonly felt as lumps in the neck, underarm, or groin areas.

### *T-cell ALL*

T-cell ALL is a less common form of ALL. T cells are lymphocytes that help fight infections. When these cells develop gene changes or an abnormal number of chromosomes, they can become cancer. These T cells lose the ability to fight infection and can multiply very quickly. This type of leukemia is more common in older children.

T-cell ALL, like B-cell ALL, can cause a large spleen or lymph nodes. Unlike B cells, T cells mature in a small immune organ located in front of the windpipe, called the thymus. When T cells develop into leukemia, the thymus can also enlarge, which can sometimes lead to breathing problems.

For more detailed information, see Prognostic Factors and Survival Rates for Childhood Leukemia.

For more detailed information on the treatment of ALL, see Childhood Acute Lymphoblastic Leukemia (ALL) Treatment.

## Acute myeloid leukemia (AML)

AML is typically a fast-growing cancer that starts in one of the following types of early (immature) bone marrow cells:

- **Myeloblasts:** These cells normally become white blood cells called granulocytes (neutrophils, eosinophils, and basophils).
- **Monoblasts:** These cells normally become white blood cells called monocytes and macrophages.
- **Erythroblasts:** These cells normally become red blood cells.
- **Megakaryoblasts:** These cells normally become megakaryocytes, the cells that make platelets.

AML has many subtypes, which are based mainly on:

- If the leukemia cells look most like cells that were meant to become white blood cells, red blood cells, or platelets and how mature they were before becoming cancer (**immunophenotype**)
- If the leukemia cells have certain **gene or chromosome changes**
- If the leukemia is related to the **treatment of an earlier cancer** (with chemotherapy or radiation)
- If the child with leukemia has **Down syndrome**

Children with AML may experience symptoms like low energy, fevers, weight loss, and enlarged lymph nodes, spleen or liver. In some cases, children with AML can experience headaches, confusion, or difficulty breathing. This may happen when there are too many myeloblasts in the bloodstream, which can slow or block blood flow.

For more detailed information see Prognostic Factors and Survival Rates for Childhood Leukemia.

For more detailed information on the treatment of AML, see Childhood Acute Myeloid Leukemia (AML) Treatment.

### Chronic leukemias

Chronic leukemias are rare in children. These leukemias tend to grow more slowly than acute leukemias. There are two main types of chronic leukemias in children, chronic lymphocytic leukemia (CLL) and chronic myeloid leukemia (CML).

## Chronic lymphocytic leukemia (CLL)

CLL is a cancer of lymphocytes and is described by stage. A stage is assigned based on blood counts and the size of the lymph nodes, spleen, or liver at diagnosis, which tells doctors how far the cancer has spread.

CLL is extremely rare in childhood. For this reason, when it occurs, it is treated similarly to adults. In people with no symptoms, because CLL is slow growing, doctors may hold off on treatment until there are signs that the cancer is progressing. If treatment is needed, targeted drugs, chemotherapy, and immunotherapy may be used for CLL. In some rare cases, radiation or stem cell transplant may be used.

For more information on adult CLL, see Chronic Lymphocytic Leukemia (CLL)[4].

## Chronic myeloid leukemia (CML)

CML is typically a slower-growing cancer of early (immature) myeloid bone marrow cells. CML accounts for only 3% of leukemias in children and is much more common among adults.

CML is caused by a specific gene change, called the *BCR-ABL* fusion or Philadelphia chromosome. It is a slow disease with 3 phases: chronic, accelerated, and blast phase. These phases are based on the number of immature blood cells seen in the blood or bone marrow. Depending on the phase at diagnosis, children may have no symptoms or have symptoms like low energy and weight loss.

CML is extremely rare in childhood. For this reason, when it occurs it is treated similarly to adults. Targeted drugs called tyrosine kinase inhibitors (TKIs) are often used to treat CML. These include drugs like imatinib (Gleevec), dasatinib (Sprycel), nilotinib (Tasigna), and bosutinib (Bosulif). Stem cell transplant may be an option if these drugs are not successful.

For more detailed information, see Prognostic Factors and Survival Rates for Childhood Leukemia[5].

For more detailed information on the treatment of CML in children, see Childhood Chronic Myeloid Leukemia Treatment.

### Juvenile myelomonocytic leukemia (JMML)

JMML is a rare type of leukemia that is neither chronic nor acute. It usually does not grow as fast as AML or as slowly as CML. It occurs most often in young children (average age of 2 years).

Symptoms can include pale skin, fever, cough, easy bruising or bleeding, trouble breathing (from too many white blood cells in the lungs), rash, and enlarged spleen, liver, and/or lymph nodes.

JMML is often caused by gene changes that affect the *RAS/MAPK* pathway. It can also be associated with Neurofibromatosis type 1[6], a family cancer syndrome, and Noonan syndrome, a rare inherited condition that affects how children grow and develop.

For more detailed information, see Prognostic Factors and Survival Rates for Childhood Leukemia.

For more detailed information on the treatment of JMML in children, see Juvenile Myelomonocytic Leukemia Treatment.

## Quick Guides

After a Leukemia Diagnosis[7]
If your child has just been diagnosed with leukemia, this short guide can help you know what to expect.

Childhood Cancer Fact Sheet[8]
Get the facts about the most common childhood cancers, including signs and symptoms, treatments, and survivorship plans.

## Hyperlinks

1. www.cancer.org/cancer/understanding-cancer/anatomy-gallery/blood-cells-bone-marrow.html
2. www.cancer.org/cancer/types/neuroblastoma.html
3. www.cancer.org/cancer/types/rhabdomyosarcoma.html
4. www.cancer.org/cancer/types/chronic-lymphocytic-leukemia.html
5. www.cancer.org/cancer/types/chronic-lymphocytic-leukemia.html
6. www.cancer.org/cancer/risk-prevention/genetics/family-cancer-syndromes/neurofibromatosis-type-1.html
7. www.cancer.org/content/dam/cancer-org/cancer-control/en/booklets-flyers/after-a-leukemia-diagnosis.pdf
8. www.cancer.org/content/dam/cancer-org/cancer-control/en/booklets-flyers/childhood-cancer-fact-sheet-patient-version.pdf
9. donate.cancer.org/?campaign=ctype
10. www.cancer.org/cancer/childhood-cancer.html
11. www.cancer.org/cancer/caregivers.html
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14. www.cancer.org/research/acs-research-highlights/childhood-cancer-research-highlights.html
15. www.cancer.org/research/acs-research-news.html?q=&tag=cancer-topics%3Acancer-in-children
16. www.cancer.org/cancer/latest-news.html?q=&tag=cancer-topics%3Acancer-in-children
17. www.cancer.org/cancer/latest-news/stories-of-hope.html?tag=cancer-topics%3Acancer-in-children

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### More Resources

Childhood Cancer[10]
Caregivers & Family[11]
Programs & Services[12]
Gold Together[13]

### News & Stories

Childhood Cancer Research Highlights[14]
Childhood Cancer Research News[15]
Latest Childhood Cancer News[16]
Childhood Cancer Stories of Hope[17]

# Key Statistics for Childhood Leukemia

Leukemia is the most common cancer in children and teens. Overall, however, childhood leukemia is a rare disease.

Acute leukemias
Chronic leukemias
Juvenile myelomonocytic leukemia (JMML)

### Acute leukemias

About 3 in 4 leukemias among children and teens are **acute lymphoblastic leukemia (ALL)**. Most of the remaining cases are **acute myeloid leukemia (AML)**.

- ALL is most common in early childhood, peaking between ages 2 and 5 years.
- ALL is slightly more common among Hispanic and White children than among African American and Asian American children, and it is more common in boys than in girls.
- AML is much more common in adults than children. When it occurs in childhood, it is more likely in the first year of life and the teenage years.

- AML occurs about equally among boys and girls of all races.

## Chronic leukemias

Chronic leukemias are rare in children. Most of these are **chronic myeloid leukemia (CML)**, which tends to occur more in teens than in younger children.

## Juvenile myelomonocytic leukemia (JMML)

**Juvenile myelomonocytic leukemia (JMML)** is rare. It usually occurs in younger children, with an average age of about 2 years. It is more common in boys than in girls.

Visit the American Cancer Society's Cancer Statistics Center[1] for more key statistics.

For statistics related to survival, see Survival Rates for Childhood Leukemias.

## Hyperlinks

1. cancerstatisticscenter.cancer.org/

### References

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# Causes, Risk Factors, and Prevention of Childhood Leukemia

The exact cause of most childhood leukemias is not known. Scientists have found some genetic and environmental risk factors, but most children with leukemia do not have any known **risk factors**. A risk factor is anything that affects a person's chance of getting a disease such as cancer.

What causes childhood leukemia?
Genetic risk factors
Environmental risk factors
Can childhood leukemia be prevented?

## What causes childhood leukemia?

Lifestyle-related risk factors for adult cancers like tobacco use, alcohol use, diet, exercise, and having excess weight are not known to play a major role in childhood cancers. Most children with leukemia have no known risk factors, so it is important to know that there is nothing these children or their parents could have done to prevent these cancers.

While the exact cause of many leukemias is unknown, certain gene changes[1] in normal bone marrow cells can cause them to multiply out of control and become leukemia cells. Certain environmental factors can also increase the risk of childhood leukemia.

Researchers continue to search for answers about why leukemia happens in children.

## Genetic risk factors

Genetic risk factors are those that are part of our DNA (the substance that makes up our genes), which controls how our bodies function. Only a small number of children inherit DNA changes from a parent that increases their risk of cancer. Most gene changes that cause childhood leukemia are acquired mutations that occur early, sometimes before birth.

In rare cases, exposure to radiation[2] and certain chemicals is known to increase the risk of cancer. However, not everyone exposed to these things will develop cancer. This is because each of us has unique genes that control how our bodies react to these types of exposures. Some people have different versions of these genes, called **gene variants**, that make them less effective. Children who inherit one of these gene variants may not be able to break down harmful chemicals well after exposure or repair damage to DNA from radiation. This can lead to a higher risk of developing leukemia over time.

### Genetic syndromes

Some genetic disorders increase a child's risk of developing several types of leukemia:

## Down syndrome (trisomy 21)

Children with Down syndrome have an extra (third) copy of chromosome 21. They are more likely to develop either acute lymphoblastic leukemia (ALL) or acute myeloid

leukemia (AML) than other children, with an overall risk of about 2% to 3%. Down syndrome has also been linked with transient leukemia (also known as transient myeloproliferative disorder [TMD or TAM]), a leukemia-like condition within the first month of life, which may resolve on its own with no or very little treatment.

## Li-Fraumeni syndrome

Li-Fraumeni syndrome[3] is a rare condition caused by a change in the *TP53* gene. People with this change have a higher risk of developing several kinds of cancer, including leukemia, bone or soft tissue sarcomas, breast cancer, adrenal gland cancer, and brain tumors.

## Neurofibromatosis type 1 (NF1)

NF1[4] is a cancer syndrome caused by a change in the *NF1* gene that causes tumors to grow on the skin, nerves, and bones. It increases the risk of developing several types of cancer, including leukemias.

## Bone marrow failure syndromes

When the bone marrow fails, the bone marrow is unable to make enough healthy blood cells and blood counts are low. The failing bone marrow can create cells with DNA that is fragile, has mutations, or is unprotected, leading to damage. This increases the risk of many types of cancer. Some examples of these bone marrow failure syndromes are **Fanconi Anemia** (FA), **Dyskeratosis congenita** (DKC), and **Shwachman-Diamond Syndrome** (SDS).

## Immune deficiency syndromes

In these conditions, the immune system does not work properly. This can lead to infections and the growth and division of damaged cells, causing cancers like leukemia. These conditions include **Ataxia-telangiectasia** (AT) and **Bloom syndrome**.

## Noonan syndrome

Noonan syndrome is a genetic condition that has been associated with JMML.

### Having a sibling with leukemia

Siblings of children with acute leukemia have a slightly increased chance of developing leukemia, but the overall risk is still low.

The risk for acute leukemias is much higher among twins. For AML and ALL that is diagnosed before one year of age, both identical twins almost always develop leukemia. If leukemia develops after one year of age or the twins are not identical, the risk for developing leukemia in the other twin is less, but still more than average.

Chronic leukemias do not seem to run in families. Having a parent who develops leukemia as an adult does not seem to raise a child's risk of leukemia either.

## Environmental risk factors

Environmental risk factors are things in our surroundings that increase the risk of getting leukemia, such as radiation, chemicals, and infections.

## Radiation exposure

Exposure to high levels of radiation is a risk factor for childhood acute leukemias. It is also the only known environmental risk factor for CML. Japanese atomic bomb survivors had a high risk of developing leukemias. If a baby is exposed to radiation before birth, there may also be an increased risk of childhood leukemia, but the extent of the risk is not clear.

The possible risks from fetal or childhood exposure to lower levels of radiation, like radiation from x-rays or CT scans, are not known. Some studies have found a slight increase in risk, while others have found no increased risk. Any risk increase is likely to be small. However, to be safe, most doctors recommend that pregnant women and children do not get these tests unless they are needed.

## Exposure to chemotherapy and certain other chemicals

Children and adults treated for other cancers with certain chemotherapy drugs have a higher risk of getting a second cancer[5], such as AML, later in life. Drugs like cyclophosphamide, doxorubicin, etoposide, and teniposide have been linked to a higher risk of leukemia. These leukemias usually develop within 10 years of treatment, and they tend to be harder to treat.

Exposure to chemicals such as benzene[6] (a solvent used in the cleaning industry and to manufacture some drugs, plastics, and dyes) may cause acute leukemia in adults and, rarely, in children. Exposure to perfluoroalkyl substances (PFAS) found in common household items like non-stick cookware and exposure to higher amounts of air pollution in early life have also been associated with childhood leukemias.

Several studies have found a possible link between childhood leukemia, specifically AML, and household exposure to pesticides either before birth or in early childhood. Some studies have also found parental exposure to paint and solvents to be associated with a higher risk of pediatric AML.

Studies of environmental exposures can be difficult to do. More research is needed to confirm these findings and provide more specific information about the possible risks.

## Exposure to infections

Viral infections have been linked to some types of childhood cancers.

The immune system plays a crucial role in fighting infections and removing injured or damaged cells. In some cases, researchers think that some childhood leukemias may result from an abnormal immune system response to an infection.

One theory suggests that childhood leukemia may develop more commonly in children exposed to certain germs (particularly viruses) later than normal. This “delayed infection” (after the first year of life) might affect the immune system in a way that allows abnormal cells to escape the system. This might help explain why some studies have found that the risk of childhood leukemia seems to be lower in children who were in daycare during their first year of life, which would have exposed them to common infections earlier.

Other researchers suggest that the movement of people around the world is resulting in exposure to new germs. Research suggests this may trigger an abnormal immune response which leads to leukemia development.

### Immune system suppression

Children who are getting intensive treatment to suppress their immune system (mainly children who have had organ transplants) have an increased risk of certain cancers, such as lymphoma and ALL.

Doctors are looking for ways to treat people without raising the risk of leukemia. For now, the benefits of treating life-threatening diseases with organ transplants must be balanced against the small chance of developing leukemia several years later.

### Can childhood leukemia be prevented?

In adults, the risk for cancers can be reduced by avoiding certain risk factors, such as smoking or exposure to hazardous chemicals in the workplace. However, there are not many avoidable risk factors for childhood leukemia. Recently, exposure to medical radiation from tests like x-rays and CT scans have been associated with a higher risk of childhood leukemia. Avoiding unnecessary tests may reduce your child's risk of developing leukemia, but sometimes these tests are important and appropriate to do.

## Hyperlinks

1. www.cancer.org/cancer/understanding-cancer/genes-and-cancer/gene-changes.html

2. www.cancer.org/cancer/risk-prevention/radiation-exposure/x-rays-gamma-rays/do-xrays-and-gamma-rays-cause-cancer.html
3. www.cancer.org/cancer/risk-prevention/genetics/family-cancer-syndromes/li-fraumeni-syndrome.html
4. www.cancer.org/cancer/risk-prevention/genetics/family-cancer-syndromes/neurofibromatosis-type-1.html
5. www.cancer.org/cancer/survivorship/long-term-health-concerns/second-cancers-in-adults/treatment-risks.html
6. www.cancer.org/cancer/risk-prevention/chemicals/benzene.html

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# Early Detection, Diagnosis, and Prognosis of Childhood Leukemia

Learn about the signs and symptoms of childhood leukemia, the tests that may be done for it, and the outlook for those with a diagnosis.

Can childhood leukemia be found early?
What are the signs and symptoms of childhood leukemia?
How is childhood leukemia diagnosed?
Survival and outlook for childhood leukemia
Questions to ask if leukemia has just been diagnosed

## Can childhood leukemia be found early?

At this time, there is no widely recommended screening test to look for leukemia before it starts to cause symptoms. Childhood leukemia is often found because a child has

signs or symptoms that prompt a visit to the doctor. The doctor then orders blood tests, which might point to leukemia as the cause. The best way to find these leukemias early is to pay attention to the possible signs and symptoms of this disease.

**For children at increased risk**

For children known to be at increased risk of leukemia (because of a genetic condition such as Li-Fraumeni syndrome[1] or Down syndrome, for example), most doctors recommend careful, regular medical visits to watch for signs or symptoms of leukemia.

Regular medical care is also important for other children at higher risk of leukemia, such as people who have been treated with chemotherapy, radiation therapy, or who are taking immune system-suppressing drugs. The risk of leukemia in these children, although higher than in the general population, is small.

## What are the signs and symptoms of childhood leukemia?

Leukemia symptoms can vary and often result from leukemia cells crowding out healthy blood cells. Many of these symptoms can also have other, more common causes. If your child has any of the following symptoms, it is important to see a doctor to find out the cause.

Signs and symptoms can include:

- Low energy, weakness, or pale skin (from low red blood cells)
- Frequent infections or fevers (from low normal white blood cells)
- Easy bruising or bleeding (from low platelets)
- Pain in the bones or joints
- Swelling in the belly (from a large spleen or liver)
- Enlarged lymph nodes in the neck, armpits, above the collarbone, or groin
- Weight loss and low appetite
- Coughing or trouble breathing

Most of these symptoms are likely due to other causes, but checking with a doctor is important for your child's health.

Signs and Symptoms of Childhood Leukemia
Childhood leukemia may cause certain signs and symptoms. Learn what to watch for here.

## How is childhood leukemia diagnosed?

Diagnosing childhood leukemia often involves blood tests, a bone marrow biopsy, and sometimes a spinal tap (a procedure that collects spinal fluid). If leukemia is found, more detailed tests will be done on the blood, bone marrow, and spinal fluid to learn more about the leukemia cells and how best to treat them.

These tests are also used to help track how well treatment is working.

Tests for Childhood Leukemia
In case of symptoms or an abnormal test, more testing can help find out if it's cancer. Learn about childhood leukemia diagnosis tests here.

## Survival and outlook for childhood leukemia

After a leukemia diagnosis, many people want to understand what to expect. While every child’s experience is different, survival rates can offer a general picture based on past outcomes in children with similar types of leukemia.

Survival rates are a way to measure how many people survive a certain type of cancer over time. Survival rates are often reported as 5-year survival rates, which refer to the percentage of children who live at least 5 years after their cancer is diagnosed. Of course, often children live much longer than 5 years (and many are cured). Some people find this information helpful, while others prefer to focus more on the treatment plan and next steps.

A child’s outlook depends on many factors, including the type of leukemia they have, how well it responds to treatment, and certain features of the leukemia cells. These are known as prognostic factors, and they help doctors tailor treatment and better predict the likely course of the disease.

Prognostic Factors and Survival Rates for Childhood Leukemia

In children with cancer, certain factors that can affect a child's outlook are called prognostic factors. Survival rates of childhood leukemia are based on outcomes of people who've had the disease. Find the survival rates and prognostic factors for childhood leukemia here.

## Questions to ask if leukemia has just been diagnosed

It is important to have open, honest discussions with your child's cancer care team. They want to answer all your questions, no matter how small they might seem. Here are some questions you might ask:

- What type of leukemia does my child have?
- How will the type of leukemia or any other factors affect my child's prognosis?
- Do we need other tests before we can decide on treatment?
- Will we need to see other doctors?
- How much experience do you have treating this type of leukemia?
- Who else will be on the treatment team, and what do they do?

## Hyperlinks

1. www.cancer.org/cancer/risk-prevention/genetics/family-cancer-syndromes/li-fraumeni-syndrome.html

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# Signs and Symptoms of Childhood Leukemia

Many of the symptoms of childhood leukemia are not specific to leukemia and could be caused by other things. If your child has any of the symptoms below, it is important to have your child seen by a doctor so the cause can be found and treated, if needed.

Symptoms from low blood counts
Symptoms from leukemic blasts in the bloodstream
Symptoms from leukemic blasts in the body

## Symptoms from low blood counts

Leukemia begins in the bone marrow, which is where new blood cells are made. As leukemia cells multiply in the marrow, they can crowd out the normal blood cells. When this happens, the bone marrow can have a tough time making enough normal red blood cells, white blood cells, and platelets. These shortages can cause symptoms. The leukemia cells can also spill into the bloodstream going to other parts of the body, which can also cause symptoms.

## Low red blood cell counts (anemia)

Red blood cells carry oxygen to all the cells in the body. A shortage of red blood cells can cause symptoms such as:

- Fatigue (tiredness)
- Weakness
- Dizziness or lightheadedness
- Shortness of breath
- Pale skin

Symptoms from anemia may be treated with a blood transfusion.

## Symptoms from low platelet counts (thrombocytopenia)

Platelets in the blood normally help stop bleeding. A shortage of platelets can lead to:

- Easy bruising (or small red or purple dots, called petechiae) and bleeding
- Frequent or severe nosebleeds
- Heavy menstrual bleeding in young women
- Bleeding gums

Bleeding from low platelets can be treated with a platelet transfusion.

## Symptoms from low or abnormal white blood cells

Normal white blood cells help the body fight off germs. Children with leukemia can have high or low white blood cell counts. The white blood cells in a person with leukemia may not function well. This can lead to:

- **Frequent infections** or infections that do not seem to go away
- **Fevers,** which are caused by infection or inflammation in the body as a response to leukemia cells, without an infection

Infections in children with leukemia can be serious. Fevers are often treated with strong antibiotics and blood cultures are taken to check for bacteria.

## Symptoms from leukemic blasts in the bloodstream

Leukemia white blood cells, called **blasts**, are bigger than normal white blood cells and are not able to fight infections like normal white blood cells. Large amounts of leukemic blasts in the blood can cause **leukostasis** and **tumor lysis syndrome** (TLS). Symptoms from leukostasis and TLS do not happen in all cases of leukemia. However, when they do, they may require urgent treatment.

## Leukostasis

**Leukostasis** is slow or blocked blood flow through small blood vessels because of the high number and large size of blasts. This can cause symptoms of a stroke, like confusion, weakness or numbness, or other symptoms like chest pain or trouble breathing.

Very high white blood cell counts that cause leukostasis can be treated by giving low doses of chemotherapy to clear some of the white blood cells, or with a process called leukapheresis, where a machine filters the white blood cells out of the bloodstream. This requires a catheter to be placed in a large vein, usually in the neck or groin.

## Tumor lysis syndrome

**Tumor lysis syndrome (TLS)** can occur when high numbers of blasts break down in a short period of time and can cause an imbalance in the salts of the blood, like potassium, uric acid, and phosphate. These and other substances from the blasts can build up in the blood. TLS can cause kidney and heart problems if it is not recognized and treated promptly.

TLS is often prevented or managed with fluids and medications. In serious cases, where the kidneys are damaged and salts in the blood build up, kidney dialysis may be needed. This is where a machine does the job of the kidney while it is damaged, removing the extra salts and chemicals. In most cases, if dialysis is needed, it is only for a few days to weeks.

## Symptoms from leukemic blasts in the body

Leukemic blasts in the body can cause a variety of symptoms.

### Bone or joint pain

This pain is caused by the buildup of leukemia cells in the bone marrow causing stress on the bone. Some children may limp or refuse to walk because of pain in their bones.

### Loss of appetite and weight loss

If the spleen and/or liver get big enough, they can press against other organs, like the stomach. This can make the child feel full after eating only a small amount of food, leading to a loss of appetite and weight loss over time.

### Swelling of the abdomen (belly)

Leukemia cells can collect in the liver and spleen, making these organs bigger. This might be noticed as fullness or swelling of the belly. The lower ribs usually cover these organs, but when they are enlarged the doctor can often feel them.

### Swollen lymph nodes

Sometimes leukemia cells spread to the lymph nodes, which are normally small (bean-sized) collections of immune cells in the body. Swollen nodes may be seen or felt as lumps under the skin in certain parts of the body (such as on the neck, in the underarm areas, above the collarbone, or in the groin). Lymph nodes inside the chest or abdomen can also swell, but these can only be seen on imaging tests[1], such as CT or MRI scans.

In infants and children, lymph nodes often get bigger when they are fighting an infection. An enlarged lymph node in a child is much more often a sign of infection than leukemia, but it should be checked by a doctor and followed closely.

Finding lymph nodes above the collarbone or lymph nodes in multiple areas at the same time is more worrisome than lymph nodes in one area like the neck or underarm. Lymph nodes that continue to get bigger or do not reduce in size within days to weeks should be seen by a doctor to find a cause and get treatment, if needed.

## Swelling of the face and arms

An enlarged thymus or lymph nodes in the chest can press on the superior vena cava (SVC), which is a large vein that carries blood from the head and arms back to the heart. This can block the blood from flowing back to the heart. This is known as **SVC syndrome**. It can result in swelling in the face, neck, arms, and upper chest (sometimes with a bluish-red skin color). Symptoms can also include headaches, dizziness, and a change in consciousness if it affects the brain. SVC syndrome can be life-threatening, so it needs to be treated right away.

## Swelling of the testicles

Rarely, leukemia in male children can cause one or both testicles to be larger than normal. It is rare for this to happen when leukemia is first diagnosed and is more likely to happen if leukemia returns after treatment.

Most of the symptoms above are more likely to be caused by something other than leukemia. Still, it is important to have these symptoms checked by a doctor so the cause can be found and treated, if needed.

## Coughing or trouble breathing

Some types of leukemia can affect structures in the middle of the chest, such as lymph nodes or the thymus (a small organ in front of the trachea, the breathing tube that leads to the lungs). An enlarged thymus or lymph nodes in the chest can press on the trachea, causing a cough or trouble breathing. In many cases this can be life-threatening, so it needs to be treated right away.

### Headaches, seizures, vomiting

A small number of children have leukemia that has already spread to the brain and spinal cord when it is first found. This can lead to symptoms such as headaches, trouble concentrating, weakness, seizures, vomiting, problems with balance, and blurred vision.

### Rashes or gum problems

In children with acute myeloid leukemia (AML), leukemia cells may spread to the gums, causing swelling, pain, and bleeding.

If AML spreads to the skin, it can cause a red or purple rash. A collection of AML cells under the skin or in other parts of the body can form a lump called a **chloroma** or **granulocytic sarcoma**. Juvenile myelomonocytic leukemia (JMML) may also cause skin changes.

## Hyperlinks

1. www.cancer.org/cancer/diagnosis-staging/tests/imaging-tests/imaging-radiology-tests-for-cancer.html

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# Tests for Childhood Leukemia

Most of the signs and symptoms of childhood leukemia are more likely to have other causes, such as an infection. Still, it is important to let your child's doctor know about such symptoms right away so that the cause can be found and treated, if needed.

Exams and tests will be done to determine the cause of symptoms. If leukemia is found, further tests will be needed to find out the type of leukemia and decide how best to treat it.

Medical history and physical exam
Tests to look for leukemia in children
Lab tests to diagnose and classify leukemia
Imaging tests

## Medical history and physical exam

If your child has signs and symptoms that suggest they might have leukemia, the doctor will ask about the symptoms and how long your child has had them. The doctor may also ask about exposure to possible risk factors. A family history of cancer, especially leukemia, may also be important.

The doctor will examine your child for possible signs of leukemia or other health problems. The exam may include vital signs (temperature, heart rate, and blood pressure) and looking at the skin, eyes and mouth for any abnormal findings like rashes, bruises, bleeding, or signs of an infection. Doctors will likely feel the abdomen (belly) for a large liver or spleen and feel for enlarged lymph nodes in the neck, underarms, and groin area. For males, a testicular exam may be done to check for swelling. Doctors may also ask a child to walk, answer questions, and test the feeling and strength of their muscles as part of a neurologic examination.

## Tests to look for leukemia in children

If the doctor thinks your child might have leukemia, blood and bone marrow samples will need to be checked to be sure. Your child's doctor may refer you to a **pediatric oncologist**, a doctor who specializes in childhood cancers (including leukemias), to have some of these tests done. If leukemia is found, other types of tests may also be done to help guide treatment.

## Blood tests

The first tests done to look for leukemia are blood tests. The blood samples are often taken from a vein in the arm, but in infants and younger children they may be taken from other veins (such as in the foot or scalp) or from a “fingerstick.”

**Blood counts** and **blood smears** are the usual tests done on these samples. A complete blood count[1] (CBC) is done to determine how many blood cells of each type are in the blood. For a blood smear, a small sample of blood is spread on a glass slide and looked at under a microscope. Abnormal numbers of blood cells and changes in the way these cells look may make the doctor suspect leukemia.

When doctors look at the blood smear of children with leukemia, they may see abnormal white blood cells and not enough red blood cells and/or platelets. White blood cells in the blood may be **blasts**, an early type of blood cell normally found only in the bone marrow. In some cases, a blood smear only shows too few or too many white blood cells. Even though these findings may make a doctor suspect that a child has leukemia, they may not be able to be diagnosed for sure without looking at a sample of bone marrow cells.

## Bone marrow aspiration and biopsy

Testing of the bone marrow includes a bone marrow aspiration and biopsy. These tests are helpful to diagnose leukemia and identify the leukemia type. A bone marrow aspiration and bone marrow biopsy are usually done at the same time. The samples are usually taken from the hip bones.

Before the tests, the skin over the hip bone is cleaned and numbed by injecting a local anesthetic or applying a numbing cream. In most cases, the child is also given other medicines to make them drowsy or go to sleep during the tests.

For a bone marrow **aspiration**, a thin, hollow needle is inserted into the bone and a syringe is used to suck out a small amount of liquid bone marrow.

For a bone marrow **biopsy**, a small piece of bone and marrow is removed by placing a slightly larger needle in the bone. Once the biopsy is done, pressure is applied to the site to help stop any bleeding.

The bone marrow samples are then sent to a lab for other tests and looked at under a microscope.

Bone marrow tests may also be repeated after a leukemia diagnosis to find out if treatment is working.

## Lumbar puncture (spinal tap)

This test is used to look for cancer cells in the cerebrospinal fluid (CSF), which is the liquid that bathes the brain and spinal cord.

It is important for the child to be still for the procedure. The doctor may do the test at the same time as other tests where the child will be under general anesthesia (where the child is asleep). If the test is done by itself, the doctor may recommend that the child be given something to make them sleep so the spinal tap can be done with minimal difficulty or chance of causing harm. Some older children may be able to have the test done while awake.

After cleaning the skin, the doctor will numb an area in the lower part of the back near the spine. A small, hollow needle is inserted between the bones of the spine to remove some of the fluid, which is then sent to the lab for testing.

In children already diagnosed with leukemia, lumbar punctures might also be used to give chemotherapy drugs into the CSF to try to prevent or treat the spread of leukemia to the spinal cord and brain. This is known as **intrathecal chemotherapy**.

## Lymph node biopsy

Leukemia in children is most often diagnosed based on blood or bone marrow results. In rare cases, a large lymph node may be the main symptom of a new leukemia. In these cases, a lymph node biopsy may be done to make a diagnosis.

During this procedure, a surgeon cuts through the skin to remove an entire lymph node (known as an excisional biopsy). If the node is near the skin surface, this is a simple operation. It is more complex if the node is inside the chest or abdomen. Most often, the child will need general anesthesia (where the child is asleep).

## Lab tests to diagnose and classify leukemia

All blood, bone marrow, and other samples are sent to a lab for further testing.

## Microscopic exams

All the samples taken (blood, bone marrow, lymph node tissue, or CSF) are looked at with a microscope. The samples might be exposed to chemical stains (dyes) that can cause color changes in some types of leukemia cells.

Doctors will look at the size, shape, and staining patterns of the blood cells in the samples to describe the type of leukemia.

A key element is whether the cells look mature or immature (look like normal blood cells or not). The most immature cells are called **blasts**. Having too many blasts, especially in the blood sample, is a common sign of acute leukemia.

An important feature of a bone marrow sample is its **cellularity**. Normal bone marrow contains a certain number of blood-forming cells and fat cells. Marrow with too many blood-forming cells is said to be **hypercellular**. If too few blood-forming cells are found, the marrow is called **hypocellular**.

## Flow cytometry and immunohistochemistry

These tests are used to determine the type of leukemia based on certain proteins in or on the cancer cells (known as **immunophenotyping**). It is most often done on cells from bone marrow, but it can also be done on cells from the blood, lymph nodes, and other body fluids.

For both flow cytometry and immunohistochemistry, samples of cells are treated with antibodies, which are proteins that stick to other proteins, called **antigens**, on cells. For immunohistochemistry, the cells are then examined under a microscope to see if the antibodies stuck to them (meaning they have these proteins), while flow cytometry uses a special machine to look for these proteins.

Flow cytometry can also be used to estimate the amount of DNA in the leukemia cells. This is important to know, especially in ALL, because cells with more DNA than normal are often more sensitive to chemotherapy and these leukemias have a better prognosis (outlook).

Flow cytometry can also be used to measure the response to treatment and the existence of minimal residual disease (MRD), or very small amounts of cancer cells, in some types of leukemias. (See Prognostic Factors and Survival Rates for Childhood Leukemias.)

## Chromosome tests

These tests look at the chromosomes (long strands of DNA) inside the cells. Normal human cells have 23 pairs of chromosomes, each of which is a certain size and looks a certain way under the microscope. But in some types of leukemia, the cells have changes in the structure and/or number of chromosomes inside.

For instance, sometimes 2 chromosomes swap some of their DNA, so that part of one chromosome becomes attached to part of a different chromosome. This structural change, called a *translocation*, can usually be seen under a microscope. Other types of chromosome changes are also possible. Recognizing these changes can help identify the exact type of leukemia and can help determine the right treatment and prognosis (outlook).

Sometimes the leukemia cells have an abnormal number of chromosomes (instead of the usual 46) – they may be missing some chromosomes or have extra copies of some. This can also affect a child's outlook.

Finding these types of chromosome changes with lab tests can be very helpful in choosing the right treatment and predicting how well the leukemia may respond to certain treatments.

For more information, see Cancer-related Genomic Testing and Genetic Testing[2].

### *Cytogenetics*

For this test, leukemia cells are grown in a lab dish and the chromosomes are looked at under a microscope. Doctors look at the number and structure of the chromosomes to find any changes.

Cytogenetic testing usually takes a few weeks because the leukemia cells take time to grow before the chromosomes are ready to be looked at.

Not all chromosome changes can be seen under a microscope. Other lab tests can often help detect these changes.

#### *Fluorescent in situ hybridization (FISH)*

This is another way to look at chromosomes and genes. It uses special dyes that only attach to specific parts of certain chromosomes. FISH can find most chromosome changes (such as translocations) that are visible under a microscope in standard cytogenetic tests, as well as some changes too small to be seen with routine cytogenetic testing.

FISH can be used to look for specific changes in chromosomes. It can be used on blood or bone marrow samples. It is very accurate and can usually provide results within a couple of days.

#### *Polymerase chain reaction (PCR)*

This is a very sensitive test that can also find chromosome and gene changes too small to be seen under a microscope, even if there are very few leukemia cells in a sample. This test can be very useful in looking for small numbers of leukemia cells especially during or after treatment.

#### *Next generation sequencing (NGS)*

This test may also be used to look for gene changes in leukemia cells that might not be seen on routine cytogenetic tests. NGS can tell us more about the best treatment for the leukemia and prognosis (outlook).

### Other blood tests

If leukemia is suspected or has been found, your child's doctor will probably order more blood tests, called a **chemistry panel**, to check liver and kidney function and the balance of salts (electrolytes) in the body. They may also check tests to make sure the blood is clotting properly (coagulation tests) and blood tests for substances that may be high when leukemia cells are present (called lactate dehydrogenase, or LDH, and uric acid).

## Imaging tests

Imaging tests use x-rays, sound waves, magnetic fields, or radioactive substances to make pictures of the inside of the body. Leukemia does not usually form large tumors, so imaging tests are not used as often as they are for other types of cancer. But if leukemia is suspected or has been diagnosed, your child's doctor may order some of these tests to get a better idea of the extent of the disease or to look for other problems, such as infections. For more details, see Imaging Tests[3].

## Chest x-rays

A chest x-ray[4] can help detect an enlarged thymus or lymph nodes in the chest. If the test result is abnormal, a computed tomography (CT) scan of the chest may be done to get a more detailed view.

Chest x-rays can also help look for pneumonia if your child has symptoms of a lung infection like cough, shortness of breath, or low oxygen levels.

## Computed tomography (CT or CAT) scan

CT scans[5] combine many x-ray pictures to make detailed cross-sectional images of the inside of the body. This test is not usually needed for children with leukemia, but it might be done if the doctor suspects the leukemia is growing in lymph nodes in the chest or in organs like the spleen or liver. It is also sometimes used to look at the brain and spinal cord, but an MRI scan may also be used for this. In rare cases where radiation is needed to treat leukemia, CT scans may be used for radiation planning.

### *PET/CT scan*

Some machines combine the CT scan with a positron emission tomography (PET) scan [6]. A PET scan uses a radioactive substance (usually a type of sugar called FDG) to mark active cells in the body. Because cancer cells multiply quickly, they absorb large amounts of the sugar and show up on the PET scan. This can provide more information about any abnormal areas that appear on the CT scan.

### Magnetic resonance imaging (MRI)

An MRI scan[7], like a CT scan, makes detailed images of soft tissues in the body. It's most helpful in looking at the brain and spinal cord, so it's most likely to be done if the doctor has reason to think the leukemia might have spread there (such as if a child has symptoms like headaches, seizures, or vomiting). This test does not use radiation. A contrast material called gadolinium may be injected into a vein before the scan to better see details. In rare cases where radiation is needed to treat leukemia, MRI scans may be used for radiation planning.

### Ultrasound (sonogram)

Ultrasound[8] uses sound waves and their echoes to look inside the body. It can be used to look at lymph nodes near the skin surface or to look for enlarged organs inside the abdomen such as the kidneys, liver, and spleen. It cannot be used to look at organs or lymph nodes in the chest because the ribs block the sound waves. There is no radiation with an ultrasound test.

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# Prognostic Factors and Survival Rates for Childhood Leukemia

Survival rates are ways to measure how many people survive a certain type of cancer over time. They might be informed by prognostic factors. They cannot tell you for sure if treatment will be successful, but they may help give you a better understanding of how likely this is. Some people find survival rates and prognostic factors helpful, but some people might not.

Understanding prognostic factors and survival rates
Acute lymphoblastic leukemia (ALL)
Prognostic factors for ALL
Risk groups and survival rates for ALL
Acute myelogenous leukemia (AML)
Prognostic factors for AML
Risk groups and survival rates for AML
Chronic myeloid leukemia (CML)
Prognostic factors for CML
Survival rates for CML
Juvenile myelomonocytic leukemia (JMML)
Prognostic factors for JMML
Survival rates for JMML

## Understanding prognostic factors and survival rates

### Prognostic factors

In children with cancer, certain factors that can affect a child's outlook (prognosis) are called **prognostic factors**. They help doctors anticipate how a child's cancer may respond to different treatments, which helps them decide how intense treatment should be. Prognostic factors are better understood for acute lymphoblastic leukemia (ALL) than other types of childhood leukemia. Factors may have to do with certain features of

the child, such as their age and overall health. Factors can also be features of the leukemia, such as the white blood cell count, type of leukemia, or presence of specific gene changes in the leukemia cells.

### Survival rates

The **5-year survival rate** refers to the percentage of children who live **at least** 5 years after their leukemia is diagnosed. With acute leukemias (ALL or AML), children who are free of the disease after 5 years are likely to be cured.

Survival rates are different for each type of childhood leukemia. Knowing the type of leukemia you or your child has is important in estimating a child's outlook. Survival rates are, at best, rough estimates. Your child's doctor can tell you how these numbers apply in your case.

Current 5-year survival rates are based on children first diagnosed and treated more than 5 years ago. Improvements in treatment since then might result in a better outlook for children diagnosed today.

Accurate survival rates can be hard to find for less common forms of childhood leukemia.

## Acute lymphoblastic leukemia (ALL)

Children with ALL are often put into **risk groups** (such as low risk, standard risk, high risk, or very high risk). These risk groups are defined by factors like the child's age, blood counts at diagnosis, gene changes in the leukemia cell and how quickly cancer responds to treatment.

The 5-year survival rate for children with ALL has increased over time and is now about 90% overall. Typically, higher risk ALL needs more intense treatments for the best chance of cure. Children with low-risk ALL have a better outlook and tend to need less treatment. It is important to know that even children in higher risk groups can be cured most of the time.

## Prognostic factors for ALL

Prognostic factors can be used to assign a leukemia risk group. While all the following are prognostic factors, only certain ones are used to determine which risk group a child is in. The first 2 factors—age at diagnosis and initial white blood cell count—are thought to be the most important.

## Age at diagnosis

Children ages 1 to 10 years with B-cell ALL tend to have better cure rates. Children younger than 1 year and children 10 years or older are considered to have high-risk ALL. The outlook in T-cell ALL is not affected much by age.

## Initial white blood cell count

Children with ALL who have very high white blood cell counts (greater than 50,000 cells per cubic millimeter) when they are diagnosed are at higher risk and need more intensive treatment.

## ALL lineage

Whether a child has B-cell or T-cell leukemia does not affect the overall outlook much for children today. However, some forms of ALL need more intense treatment for the best prognosis. T-cell ALL and Burkitt leukemia (a mature B-cell leukemia) tend to need more intense treatment than other forms of ALL to achieve and maintain remission (when there are no cancer cells in the blood or bone marrow).

### Sex

Girls with ALL may have a slightly higher chance of being cured than boys, but as treatments have improved in recent years, this difference has shrunk.

### Number of chromosomes in the leukemia cells (ploidy)

Normal human cells have 46 chromosomes. Children are more likely to be cured if their leukemia cells have more than 50 chromosomes (called **hyperdiploidy**), especially if there is an extra chromosome 4 or 10. Hyperdiploidy can also be expressed as a DNA index of more than 1.16. Children whose leukemia cells have fewer than 44 chromosomes (known as **hypodiploidy**) have a less favorable outlook.

### Chromosome changes (such as translocations)

Translocations occur when chromosomes swap some of their DNA. Children whose leukemia cells have a translocation between chromosomes 12 and 21 are more likely to be cured. Those with a translocation between chromosomes 9 and 22 (the Philadelphia chromosome) or 4 and 11 tend to have a less favorable outlook. Some of these “poor” prognostic factors have become less important in recent years as treatment has improved.

### Response to initial treatment

Children whose leukemia goes into **remission** (major reduction of cancer cells in the bone marrow) within 1 to 2 weeks of chemotherapy have a better outlook than those whose leukemia does not. Having **minimal residual disease** (MRD), which is a very

small amount of leukemia cells still detectable by sensitive lab tests, can also affect outlook. Children whose cancer does not respond well may be given more intensive chemotherapy.

## Risk groups and survival rates for ALL

For a child with ALL, the risk group is important in estimating their outlook. Risk groups may vary slightly between treatment protocols and clinical trials.

The overall 5-year survival for children with ALL is about 90%. Here are general survival statistics based on risk groups. These numbers come from past clinical trials treating children with ALL.

## Low-risk group

Children with low-risk ALL:

- Are between ages 1 and 10 years
- Have white blood cell counts of less than 50,000 when they are diagnosed
- Have favorable gene changes and responded well to treatment early after diagnosis (within the first 1 to 2 weeks).

Initial treatment is successful for children in the low-risk group more than 95% of the time. If leukemia recurs, there are other options to treat and hopefully still cure the leukemia.

## Average-risk group

Children with average-risk ALL:

- Are ages 1 to 10 years old
- Have white blood cell counts of less than 50,000 when they are diagnosed

- Either respond quickly to treatment, or have favorable gene changes and respond a little slower but have no cancer cells at the end of induction treatment

For those in the average-risk group, initial treatment is successful about 90% to 95% of the time. A recent study adding an immunotherapy drug, called blinatumomab, to chemotherapy showed even better outcomes for children with average-risk ALL. If leukemia recurs, there are other options to treat, and hopefully still cure, the leukemia.

## High-risk group

The high-risk ALL group consists of children:

- Ages 10 years or older or with white blood cell counts over 50,000 when they are diagnosed and that responded well to treatment.
- Ages 1 to 10 years with white blood cell counts of less than 50,000 who responded slowly to treatment, but still have no cancer cells on standard tests at the end of induction treatment (the first 4 weeks of treatment)
- With leukemia in the cerebrospinal fluid (CSF) or testicles at diagnosis.

ALL can be classified as high risk (instead of very high risk) in these children even when there is still a very small amount of cancer cells in the body after induction found on special, more sensitive, testing called minimal residual disease (MRD).

For those in the high-risk group, initial treatment is successful in 88% to about 90% of cases. Blinatumomab is now being used in high risk ALL too, which may improve these outcomes. If leukemia recurs, there are other options to treat and hopefully still cure the leukemia.

## Very high-risk group

Very high-risk ALL is ALL that does not respond well to treatment, meaning the cancer does not respond or go away during the first 4 weeks of treatment (induction), or has unfavorable gene changes (such as KMT2A).

Infants, or children less than 1 year old at the time of diagnosis are also harder to treat and in the very high-risk group. Blinatumomab is being added in newer trials to treat infants, with improved outcomes. There may be open clinical trials specifically for infants to test new treatments and strategies.

For those in the very high-risk group, the risk of cancer coming back is more common than in other risk groups. Initial treatments are still successful in many cases, but more intense treatments such as stem cell transplant (also known as a bone marrow transplant) may be needed.

## Special groups: T-cell ALL, Early T-cell precursor ALL, Philadelphia-chromosome-positive (Ph+) and Ph-like ALL

### *T-cell ALL*

In the past, T-cell ALL was much harder to treat than B-cell ALL. However, with new treatments the outlook for T-cell ALL has improved. Survival rates for T-cell ALL range from 85-90% with intensive treatment today.

### *ETP ALL*

ETP ALL is a type of T-cell ALL. In the past, ETP ALL was thought to have a worse outlook. While ETP ALL tends to respond to treatment slower, overall prognosis for ETP ALL is similar to T-cell ALL. ETP ALL is treated the same way as T-cell ALL.

### *Ph+ and Ph-like ALL*

B-cell ALL with gene changes called the Philadelphia chromosome, a *BCR-ABL* translocation, tend to be harder to treat. Targeted drugs called tyrosine-kinase inhibitors (TKI) are used in Ph+ and Ph-like leukemia. With TKI treatment, Ph+ ALL survival rates range from 86-89%.

Ph-like leukemia remains harder to treat, with a worse outlook. Younger children with Ph-like ALL do better than older children and young adults. Survival rates range from 25-60% based on age and specific gene changes. For some types of Ph-like leukemia, targeted drugs may be used to improve outlook. If your child has Ph-like ALL, talk to the cancer care team about the outlook in your child's case.

## Acute myelogenous leukemia (AML)

The overall 5-year survival rate for children with AML has also increased over time and is now in the range of 65-70%. However, survival rates vary depending on the subtype of AML and other factors.

## Prognostic factors for AML

Prognostic factors are not quite as important in predicting outcome or in guiding treatment for AML as they are for ALL.

## Age at diagnosis

Infants and children newly diagnosed with AML tend to have a better outlook than adolescents and young adults. Adolescents and young adults, aged 15 to 39, may be more likely to relapse (have cancer return after treatment) and have more side effects from treatment.

## Initial white blood cell count

Children with AML whose white blood cell count is less than 100,000 cells per cubic millimeter at diagnosis tend to do better than those with higher counts.

## AML type

Some types of AML tend to have a better outlook than others. For example, acute promyelocytic leukemia tends to have a better outlook than most other types of AML.

## Chromosome or gene changes

Specific gene or chromosome changes are important prognostic factors in AML. Researchers agree on low-risk gene changes in AML. There is some debate about other gene changes and whether they are adverse/high-risk or neutral—not having a major impact on outlook and treatment.

Favorable or low-risk gene changes in AML include:

- Translocations, or swapping of genes, between chromosome 8 and 21 [also noted as t(8;21), or *RUNX1-RUNX1T1*]
- An inversion in chromosome 16
- *CEBPA* gene changes
- *NPM1* gene changes without *FLT3-ITD*

Some of the more commonly acknowledged adverse or high-risk gene changes in AML include:

- Missing a copy of chromosome 5 or 7, called monosomy 5 or monosomy 7
- Translocation of a specific part of chromosome 11, that involves a gene called *KMT2A*, with chromosome 4, 6 or 10
- *FLT3-ITD* gene changes [also noted as t(6;9)(p23;q34)]
- An inversion in chromosome 3

## Down syndrome

Children with Down syndrome who develop AML tend to have a good outlook if the child is 4 years old or younger at the time of diagnosis.

## Myelodysplastic syndrome or secondary AML

Children who first have a myelodysplastic syndrome[1] or whose leukemia is the result of treatment for another cancer tend to have a less favorable outlook.

## Response to initial treatment

Children whose AML responds quickly to treatment (only one chemotherapy cycle needed to achieve remission) are more likely to be cured than those whose leukemia takes longer to respond or does not respond at all.

## Risk groups and survival rates for AML

For a child with AML, the risk group is important in estimating their outlook. Researchers agree about low-risk gene changes in childhood AML, however there is some debate about the impact of other gene changes on prognosis and treatment.

The overall 5-year survival rate for children with AML is about 65-70%. Here are general survival statistics based on risk groups. These numbers come from past clinical trials treating children with AML.

## Low-risk group

Children with AML that have favorable or low risk gene changes have a better outlook than those with other types of AML.

70-80% of these children will be cured with chemotherapy alone. If cancer recurs, there are other options to treat it such as stem cell transplant (also known as bone marrow transplant).

## Intermediate-risk group

Children with AML in the intermediate-risk group have neither low-risk nor high-risk gene changes. About 50-60% of people will be cured. However, the outlook in this type of AML is predicted by how well the cancer responds to the first course of treatment, called induction.

Children that have no measurable disease after induction have a better outlook than those who still have measurable disease.

### High-risk group

Children with AML that has adverse or high-risk gene changes are harder to treat. Children with high-risk AML often have survival rates of less than 50%, but survival in high-risk AML varies based on the gene change present and how well the cancer responds to early treatment with chemotherapy.

You should discuss your child's specific outlook with your child's cancer treatment team.

### Special groups: Acute promyelocytic leukemia

APL is a unique type of AML. APL can be standard risk (a white blood cell count less than 10,000 at diagnosis) or high risk (a white blood cell count of 10,000 or more). High-risk APL can be more difficult to treat early on and may have more bleeding at diagnosis, which can be severe. The risk group helps doctors decide what treatment is best.

Overall, APL has a good outlook for children with more than 95% survival with current treatments that are different than those used for treating other forms of AML.

## Chronic myeloid leukemia (CML)

CML is a rare chronic leukemia in childhood, almost never seen in children less than 4 years old. It causes about 3% of leukemia in children. It has two main phases over time, a chronic phase and a blast crisis phase.

Prognostic factors such as age, spleen size and laboratory test results are used in adults with CML to assign risk groups. These factors seem to be less predictive of outlook in children with CML.

Today, the 5-year survival rate for children with CML is more than 92%. However, many children may need to stay on treatment for many years to keep CML in remission.

## Prognostic factors for CML

### Phase at diagnosis

About 95% of CML is diagnosed in the chronic phase which can last for a few years if untreated. In rare cases, CML is first diagnosed in blast crisis phase. CML in blast crisis can have signs and symptoms like acute leukemia, such as ALL or AML and is more difficult to treat than chronic phase CML.

CML blasts that have features of lymphoblasts have a better outlook than CML blasts with myeloid features.

### Response to Tyrosine Kinase Inhibitor (TKI) therapy

For children with chronic phase CML, a good response to TKI treatment is the most important prognostic factor.

Chronic phase CML that does not respond to TKI treatment is harder to treat, and the outlook is worse than CML that responds well to TKIs. Stem cell transplant is still another treatment option for these children.

## Survival rates for CML

In the past, stem cell transplants from matched family members were the most successful treatment, curing 70-80% of children with CML. Newer drugs, called TKIs, transformed the treatment of CML in both children and adults in the early 2000s.

More than 92% of children with CML survive at least 5 years with the introduction of TKIs as part of treatment.

## Juvenile myelomonocytic leukemia (JMML)

JMML is a rare and aggressive leukemia. The overall 5-year survival rate for children with JMML is about 45-55%. Different gene changes that lead to JMML may impact the choice of treatment and outlook. Stem cell transplant with or without low dose chemotherapy before transplant may be a part of your child's treatment plan depending on these gene changes.

## Prognostic factors for JMML

### Age at diagnosis

Children younger than 2 years of age at diagnosis tend to have a better outlook in JMML.

### Blood counts

Children with platelet counts above 33,000 and/or low fetal hemoglobin levels tend to have a better outlook. Fetal hemoglobin is a protein that carries oxygen in some red blood cells in young children.

### Changes in genes and gene expression

JMML is most often caused by a change to 1 of the 5 genes in the RAS oncogene[2] pathway. These genes are *NF1*, *NRAS*, *KRAS*, *PTPN11*, and *CBL*. JMML with more than 1 gene change has a worse outlook than children with only 1 gene change.

In some cases, the genes are normal but are turned on or off by a process called methylation which changes how genes are expressed. JMML with lower levels of methylation tends to have a better prognosis.

## Genetic syndromes

JMML is more common in children with neurofibromatosis type 1[3], a family cancer syndrome, and Noonan syndrome.

Children with Noonan syndrome are short, may have a broad or webbed neck, and may have heart problems and trouble in school. Some infants with Noonan syndrome develop features of JMML in the blood, but do not meet the criteria for diagnosis. These changes can be watched closely and may resolve without treatment over time.

## Survival rates for JMML

The overall 5-year survival rate of children with JMML is about 50-75% with the use of stem cell transplant for treatment.

## Hyperlinks

1. www.cancer.org/cancer/types/myelodysplastic-syndrome.html
2. www.cancer.org/cancer/understanding-cancer/glossary.html#alpha-O
3. www.cancer.org/cancer/risk-prevention/genetics/family-cancer-syndromes/neurofibromatosis-type-1.html

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with juvenile myelomonocytic leukemia in Korea: a report of the Korean Pediatric Hematology-Oncology Group. Bone Marrow Transplant. 2023;58(1):20-29.

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# Treating Childhood Leukemia

If your child has been diagnosed with leukemia, the cancer care team will discuss treatment options with you. It is important to weigh the benefits of each treatment option against the possible risks and side effects.

How is childhood leukemia treated?
Who treats leukemia in children?
Making treatment decisions
Preparing for treatment
Social and emotional health during treatment
Help getting through cancer treatment

## How is childhood leukemia treated?

The main treatments for most childhood leukemias are medications like chemotherapy, immunotherapy, and targeted drugs. For some children with higher-risk leukemias or leukemias that have not responded well to treatment, high-dose chemotherapy may be given along with a stem cell transplant. Other treatments may be used in special circumstances.

Chemotherapy for Childhood Leukemia
Chemotherapy (chemo) is a treatment of cancer-killing drugs used to kill childhood leukemia cells. Learn more about chemotherapy here.

Targeted Therapy Drugs for Childhood Leukemia
Targeted drugs work differently than standard chemotherapy drugs. These drugs can be helpful in treating some types of childhood leukemia. Learn more here.

Immunotherapy for Childhood Leukemia
Immunotherapy drugs work by stimulating the child's immune system to fight leukemia. Learn more here.

High-dose Chemotherapy and Stem Cell Transplant for Childhood Leukemia
A stem cell transplant (SCT) allows doctors to use higher doses of chemo to kill childhood leukemia cells.

Surgery for Childhood Leukemia
Surgery has a very limited role in the treatment of childhood leukemia. Learn about when surgery might be used here.

Radiation Therapy for Childhood Leukemia
Radiation therapy is a treatment that uses high-energy rays or particles to destroy childhood leukemia cells. Learn more here.

**Common treatment approaches**

After leukemia is diagnosed and tests have been done to determine the specific type, your child's cancer care team will discuss the treatment options with you. The most important factor in choosing a treatment is the type of leukemia, but other factors also play a role.

Treatment of acute forms of childhood leukemia (ALL or AML) is usually very intensive, so it is important that it take place in a center that specializes in treating childhood cancers.

Treatment of Children with Acute Lymphocytic Leukemia (ALL)
The main treatment for children with ALL is chemotherapy, which is usually divided into 3 phases. Other treatments might be options as well. Learn more here.

Treatment of Children with Acute Myeloid Leukemia (AML)
The main treatment for most children with acute myeloid leukemia (AML) is chemotherapy, which is usually divided into 2 phases. Learn more here.

Treatment of Children with Chronic Myeloid Leukemia (CML)
Chronic myelogenous leukemia (CML)is rare in children, and is treated like it is in adults. Learn more here.

Treatment of Children with Juvenile Myelomonocytic Leukemia (JMML)
Juvenile myelomonocytic leukemia (JMML)is fairly rare, and there is no single best chemotherapy treatment for this leukemia. Learn more here.

## Who treats leukemia in children?

Children and teens with leukemia and their families have special needs that can best be met by children's cancer centers[1]. These centers have teams of specialists who know the differences between cancers in adults and those in children and teens, as well as the unique needs of younger people with cancer.

For childhood leukemias, this team is typically led by a **pediatric oncologist**, a doctor who treats cancer in children. Many other health professionals may be involved in your child's care as well, including other doctors, nurses, nurse practitioners (NPs), physician assistants (PAs), psychologists, social workers, dieticians, rehabilitation specialists, and others.

## Making treatment decisions

After leukemia is diagnosed and tests have been done to determine its type, your child's cancer care team will discuss the treatment options with you.

It is important to discuss your child's treatment options as well as their possible side effects with the treatment team to help make the decision that is the best fit for your child. If there is anything you do not understand, ask to have it explained.

Learn more in Talking to Your Child's Cancer Care Team[2].

### Questions to ask before childhood leukemia treatment

### Choosing a treatment

- What are our treatment choices?
- What do you recommend and why?
- Are there any clinical trials[3] we should consider?
- How soon do we need to start treatment?

- Should we get a second opinion[4]? How would we do that? Can you recommend a doctor or cancer center?
- What will our options be if the treatment does not work or if the leukemia comes back?

**What to expect during treatment**

- What should we do to be ready for treatment?
- How long will treatment last? What will it be like?
- What are the chances of curing the leukemia with this treatment?
- Will the treatment be given in the hospital, clinic or at home?
- Can my child go to school while getting treatment? How will treatment affect our daily life?

**Possible side effects and long-term effects**

- What are the risks and side effects of the treatments you recommend?
- Which side effects start shortly after treatment, and which ones might develop later on?
- Is there anything we can do to help manage or avoid side effects?
- What symptoms or side effects should we tell you about right away?
- How can we reach you or someone on your team on nights, weekends, or holidays?
- Will treatment affect my child's ability to learn, grow, and develop?
- Will treatment affect my child's future ability to have children?

**Support and resources**

- Who can we talk to if we have questions about costs, insurance coverage, or social support?
- Do you know of any support groups where we can talk to other families who have been through this?

**Other things to consider**

- If time allows, consider **getting a second opinion**[5] to feel more confident about the treatment plan you choose for you or your child.

- **Clinical trials**[6] study new treatments and may offer access to promising options not widely available. They are also how doctors learn better ways to treat cancer. Children's cancer centers often conduct many clinical trials at any one time. Many children treated at these centers take part in a clinical trial as part of their treatment. If you would like to learn more about clinical trials that might be right for you or your child, start by asking the treatment team if your clinic or hospital conducts clinical trials.
- You may hear about ways to relieve symptoms or treat your cancer such as herbs, diets, acupuncture, massage, or many others. **Integrative (holistic) methods**[7] are used with standard care, while **alternative ones** replace it. Some may help with symptoms, but many aren't proven to work and could even be harmful. Talk with the care team first to make sure anything you're considering is safe and won't interfere with your or your child's treatment.

## Preparing for treatment

Before treatment, the doctors and other members of the team will help you understand the tests that will need to be done.

Your child's cancer care team will also often include a social worker. The team's social worker will be there to support you and your family before, during and after treatment. Adjusting to a new cancer diagnosis and its treatment plan can be tough, but the cancer care team is there to help. Learn more in Helping Your Child Adjust to a Cancer Diagnosis[8].

## Social and emotional health during treatment

Social and emotional distress may come up during and after treatment. Factors such as the child's age and the extent of treatment can play a role.

Some children have emotional or psychological health issues that need to be addressed during and after treatment. Depending on their age, they may also have some problems with normal functioning and schoolwork. These types of issues can often be helped with support and encouragement. Doctors and other members of the health care team can also often recommend special support programs and services to help children after treatment. For more information, see Helping Your Child Transition from Treatment to Survivorship[9].

Many experts recommend that school-aged patients attend school as much as they can. This can help them maintain a routine and tell their friends about what is happening.

Friends can be a great source of support, but patients and parents should know that some people may have wrong ideas about cancer and fears of their own. Some cancer centers have school re-entry programs that can help in these situations. In these programs, health educators visit the school and tell students about cancer, its treatment and what changes they may notice in their classmate. They also answer any questions their class might have. (For more information, see Going to School During and After Cancer Treatment[10].)

Parents and other family members can also be affected, both emotionally and in other ways. Some common family concerns during treatment include financial stresses, traveling to and staying near the cancer center, the need to take time off from work, and the need for home schooling. Social workers and others at your treatment center can help sort through these issues.

Centers that treat many patients with leukemia may have programs to introduce new patients and their families to others who have finished their treatment. This can give them an idea of what to expect during and after treatment, which can be very important.

## Help getting through cancer treatment

Your child's cancer care team will be your first source of information and support, but there are other resources for help when you need it. Hospital- or clinic-based support services can also be an important part of cancer care. These might include nursing or social work services, financial aid, nutritional advice, rehab, or spiritual help. For

children and teens with cancer and their families, other specialists can be an important part of care as well.

The American Cancer Society also has programs and services[11]—including rides to treatment, lodging, and more—to help you get through treatment. Call our National Cancer Information Center at 1-800-227-2345 and speak with one of our trained specialists.

*The treatment information given here is not official policy of the American Cancer Society and is not intended as medical advice to replace the expertise and judgment of your cancer care team. It is intended to help you and your family make informed decisions, together with your doctor. Your doctor may have reasons for suggesting a treatment plan different from these general treatment options. Don't hesitate to ask your cancer care team any questions you may have about your treatment options.*

## Hyperlinks

1. www.cancer.org/cancer/childhood-cancer/how-to-find-childhood-cancer-treatment.html
2. www.cancer.org/cancer/childhood-cancer/talking-to-your-childs-cancer-care-team.html
3. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/clinical-trials.html
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7. www.cancer.org/cancer/supportive-care/integrative-medicine.html
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11. www.cancer.org/support-programs-and-services.html

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# Chemotherapy for Childhood Leukemia

Chemotherapy (chemo) is the main treatment for the most common childhood leukemias. Chemo is the use of drugs to kill cancer cells. Chemo can be given in a vein (IV), in a muscle, in the cerebrospinal fluid (CSF) during a lumbar puncture (called intrathecal chemo[1]), or taken by mouth.

Chemotherapy for childhood leukemia by type
Possible side effects of chemo
More information about chemotherapy

## Chemotherapy for childhood leukemia by type

The type of chemotherapy used for treatment depends on the type of childhood leukemia.

## Chemotherapy for acute lymphoblastic leukemia (ALL)

ALL is treated with combinations of several chemo drugs. Chemo is given in phases or cycles. Rotating these drugs throughout treatment and giving them together can help

reduce the chances that the cancer cells become resistant. Some periods during treatment are more intense and may cause more side effects and require time in the hospital. Other periods during treatment allow the body to recover.

ALL treatment starts with chemo to kill the leukemia cells such that cancer cells are no longer seen on tests of the blood and bone marrow. This is called putting the leukemia

in remission. After that, chemotherapy is continued for a longer period (usually 2-3 years total) to prevent the cancer from coming back.

Some of the chemo drugs used to treat ALL include:

- Anthracycline drugs (daunorubicin, doxorubicin)
- Asparaginase (PEG-L-asparaginase [pegaspargase], calaspargase [CAL-PEG], Erwinia asparaginase)
- Cyclophosphamide
- Cytarabine
- Mercaptopurine (6MP)
- Methotrexate
- Nelarabine (for T-cell ALL)
- Steroids (prednisone, dexamethasone)
- Thioguanine (6TG)
- Vincristine

## Chemotherapy for acute myeloid leukemia (AML)

AML is treated with combinations of several chemo drugs. Chemo is given in phases or cycles. Each cycle includes a period of giving chemo, followed usually by a period of rest, which allows the body to recover from the chemo.

AML is treated with chemo over a shorter period compared to ALL, usually less than a year. Chemo for AML is more intense and may require more time in the hospital during treatment.

Some of the chemo drugs used to treat AML include:

- Anthracycline drugs (daunorubicin, doxorubicin, mitoxantrone)
- Cytarabine
- Etoposide

## Chemotherapy for chronic myeloid leukemia (CML)

Chemotherapy is not typically used in the treatment of CML. Chemotherapy may be used for CML if a stem cell transplant is recommended. See High-dose Chemotherapy and Stem Cell Transplant for Childhood Leukemia.

## Chemotherapy for juvenile myelomonocytic leukemia (JMML)

JMML is rare. The best chemo drugs for JMML are not known. Some of the chemo drugs that have been shown to work against JMML include:

- Azacitidine
- Cytarabine
- Decitabine
- Etoposide
- Fludarabine
- Mercaptopurine
- Thioguanine

None of these chemo drugs have been found to work well in JMML without a stem cell transplant.

## Possible side effects of chemo

Chemo drugs affect cells that multiply fast. Cancer cells multiply quickly, but other healthy cells in the body do too, for example blood-forming cells, hair cells, and cells that make up the lining of our gut. Damage caused by these drugs to fast-growing healthy cells results in side effects. The side effects depend on the type and doses of the drugs, and the length of time they are given.

### Common side effects

These side effects can include:

- Hair loss
- Mouth sores
- Loss of appetite
- Nausea and vomiting
- Diarrhea or constipation

### Possible side effects from low blood cell counts

Chemo can damage the bone marrow, where new blood cells are made. This can lead to low blood cell counts, which can result in:

- Increased chance of infections (from having too few white blood cells, called **leukopenia**)
- Easy bruising or bleeding (from low platelets, called **thrombocytopenia**)
- Fatigue (from low red blood cells, called **anemia**)

The problems with blood cell counts are often caused by the leukemia itself at first. They might get worse during the first part of treatment because of the chemo, but they will likely improve as the leukemia cells are killed off and the normal cells in the bone marrow have space to recover.

### Possible side effects from specific chemo drugs

Some chemo drugs have other specific side effects. For example:

- Steroids can have many side effects. These include high blood pressure, high blood sugar, changes in mood, changes in appetite, or trouble sleeping.
- Vincristine can damage nerves, which can lead to numbness, tingling, or weakness in hands or feet (known as peripheral neuropathy).
- Asparaginase can increase the risk of blood clots.

Some chemo drugs can also have long-term side effects, such as effects on growth and development, effects on fertility later in life, or an increased risk of getting a second cancer (often AML). Intrathecal (IT) chemo, given directly into the cerebrospinal fluid (CSF) around the brain and spinal cord, can have its own side effects. Intrathecal chemo may cause trouble thinking or learning in some children. It may even cause seizures.

For more on this, see After Treatment for Childhood Leukemia.

### Managing side effects

Most side effects tend to go away once treatment is finished. There are often ways to reduce these side effects. For instance, drugs can be given to help prevent or reduce nausea and vomiting. Be sure to ask your child's doctor or nurse about medicines to help reduce side effects and let them know if your child has side effects so they can be managed.

## More information about chemotherapy

For more general information about how chemotherapy is used to treat cancer, see Chemotherapy[2].

To learn about some of the side effects listed here and how to manage them, see Managing Cancer-related Side Effects[3].

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/chemotherapy/getting-chemotherapy.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/chemotherapy.html
3. www.cancer.org/cancer/managing-cancer/side-effects.html

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# Targeted Therapy Drugs for Childhood Leukemia

In recent years, new drugs that target specific parts of cancer cells have been developed. These targeted drugs work differently from standard chemotherapy drugs. They can be used instead of or along with chemo in some situations, and they have side effects that are different from chemo. Some targeted drugs can be useful in childhood leukemias.

The types of targeted drugs that are used for treatment depend on the type of leukemia.

Targeted therapy drugs for acute lymphoblastic leukemia (ALL)
Targeted therapy drugs for acute myeloid leukemia (AML)
Targeted therapy drugs for chronic myeloid leukemia (CML)
More information about targeted therapy

## Targeted therapy drugs for acute lymphoblastic leukemia (ALL)

Read about the targeted therapy drugs sometimes used to treat ALL and their possible side effects.

## Antibody-drug conjugates

**Antibody-drug conjugates (ADC)** are monoclonal antibodies (man-made immune proteins) linked to chemotherapy drugs. The antibody acts like a homing device. It brings the chemo drug to the leukemia cells, where it enters the cells and kills them when they try to divide into new cells. An example of this type of targeted therapy for AML is **gemtuzumab ozogamicin (Mylotarg)**.

This drug can be used to treat some children with AML in different situations:

- It can be used as part of the initial treatment of AML, along with chemo.
- It can be given if the leukemia comes back or is no longer responding to treatment.

This drug is given as an infusion into a vein (IV).

### *Possible side effects*

The most common side effects are fever, nausea and vomiting, low levels of blood cells (with increased risks of infection, bleeding, and fatigue), swelling and sores in the mouth, constipation, rash, and headaches.

Less common but more serious side effects can include:

- Severe liver damage, including veno-occlusive disease (blockage of veins in the liver)
- Reactions during the infusion[1] (similar to an allergic reaction). Your child will likely be given medicines before each infusion to help prevent this.
- Serious or life-threatening infections, especially in people who have already had a stem cell transplant
- Changes in heart rhythm

## BCL-2 inhibitors

These drugs attack **BCL-2**, a protein that can help leukemia cells live longer. An example is **venetoclax (Venclexta)**. In children with AML, this may be used when the cancer comes back or does not respond well to initial treatment.

These drugs are given as a tablet, by mouth, usually once daily.

### *Possible side effects*

The most common side effects are fever, feeling tired or weak, low appetite, nausea, diarrhea, low blood counts, swelling, skin rash, and changes to some blood tests which may need to be watched while getting the drug.

## Menin inhibitors

In some children with ALL, leukemia cells have a type of change known as a **translocation in the *KMT2A* gene**. This causes the cells to make an abnormal KMT2A protein, which can combine with another protein called menin to help the cells grow.

**Revumenib (Revuforj)** is a menin inhibitor. It stops menin from combining with the KMT2A protein. This drug can be used to treat children with ALL whose leukemia cells have a *KMT2A* translocation, if the leukemia is no longer responding to treatment or has come back (relapsed) after other treatments. The child's blood or bone marrow can be tested to see if the leukemia cells have this gene change.

This drug is taken by mouth as pills, typically twice a day.

#### *Possible side effects*

Common side effects of this drug can include nausea, vomiting, bleeding, diarrhea or constipation, fatigue, muscle pain, increased risk of infection, swelling in the arms and legs, and loss of appetite.

This drug may cause serious heart rhythm problems. This might lead to an irregular heartbeat, which could be life threatening. Your child's doctor will check blood mineral levels and get electrocardiograms (EKGs) to test your child's heart rhythm before and during treatment with this drug.

An important possible side effect of this drug is differentiation syndrome. This occurs when the leukemia cells release certain chemicals into the blood. It most often occurs shortly after starting treatment, but sometimes it can happen weeks or months later. Symptoms can include fever, cough, breathing problems (from fluid buildup in the lungs and around the heart), dizziness or lightheadedness (from low blood pressure), urinating less often (from damage to the kidneys), and severe fluid buildup elsewhere in the body. It can often be treated by stopping the drug for a while and giving other medicines (such as dexamethasone or hydroxyurea).

### Tyrosine kinase inhibitors (TKIs)

TKIs work by blocking the signals cells need to grow and divide. Examples of TKIs include **imatinib (Gleevec), dasatinib (Sprycel), nilotinib (Tasigna), bosutinib (Bosulif)**, **ponatinib (Iclusig)** and **ruxolitinib (Jakafi)**.

A small number of children with acute lymphocytic leukemia (ALL) have the Philadelphia chromosome (**Ph+ ALL**). TKIs attack cells that have the *BCR-ABL* gene change. For these children, combining chemo and TKI drugs results in better outcomes. They may also be used in treating some children with **Ph-like ALL**, where leukemia cells have gene changes other than *BCR-ABL* that cause growth signals to be overactive.

These drugs are taken daily as pills or capsules. Imatinib is also available as a liquid, under the name Imkeldi.

#### *Possible side effects*

Possible side effects of these drugs include diarrhea, nausea, muscle pain, fatigue, and skin rashes. These are generally mild. A common side effect is swelling around the eyes or in the hands or feet, which may be caused by the drugs' effects on the heart. Other possible side effects include lower blood counts when treatment starts. These drugs might also slow a child's growth, especially if used before puberty.

## Targeted therapy drugs for acute myeloid leukemia (AML)

Read about the targeted therapy drugs sometimes used to treat AML and their possible side effects.

## Antibody-drug conjugates

**Antibody-drug conjugates (ADC)** are monoclonal antibodies (man-made immune proteins) linked to chemotherapy drugs. The antibody acts like a homing device. It brings the chemo drug to the leukemia cells, where it enters the cells and kills them when they try to divide into new cells. An example of this type of targeted therapy for AML is **gemtuzumab ozogamicin (Mylotarg)**.

This drug can be used to treat some children with AML in different situations:

- It can be used as part of the initial treatment of AML, along with chemo.
- It can be given if the leukemia comes back or is no longer responding to treatment.

This drug is given as an infusion into a vein (IV).

#### *Possible side effects*

The most common side effects are fever, nausea and vomiting, low levels of blood cells (with increased risks of infection, bleeding, and fatigue), swelling and sores in the mouth, constipation, rash, and headaches.

Less common but more serious side effects can include:

- Severe liver damage, including veno-occlusive disease (blockage of veins in the liver)
- Reactions during the infusion[2] (similar to an allergic reaction). Your child will likely be given medicines before each infusion to help prevent this.
- Serious or life-threatening infections, especially in people who have already had a stem cell transplant
- Changes in heart rhythm

### BCL-2 inhibitors

These drugs attack **BCL-2**, a protein that can help leukemia cells live longer. An example is **venetoclax (Venclexta)**. In children with AML, this may be used when the cancer comes back or does not respond well to initial treatment.

These drugs are given as a tablet, by mouth, usually once daily.

#### *Possible side effects*

The most common side effects are fever, feeling tired or weak, low appetite, nausea, diarrhea, low blood counts, swelling, skin rash, and changes to some blood tests which may need to be watched while getting the drug.

### Differentiation agents

Differentiation agents can be used to treat a special type of AML, called **acute promyelocytic leukemia (APL)**. The leukemia cells in APL (called blasts), have certain gene changes that stop them from maturing into normal white blood cells. These drugs help the blasts mature (differentiate) into normal white blood cells. Two of these drugs can be used to treat APL.

**All-trans retinoic acid (ATRA, tretinoin)** is a form of vitamin A that is typically part of the initial treatment of APL. It is given either along with chemo or along with ATO. It can also be used during later phases of treatment.

**Arsenic trioxide (ATO)** can act in a way similar to ATRA in patients with APL and kills the blast cells. It is often given with ATRA in the treatment of APL, but it is also helpful in treating APL that comes back after treatment with ATRA plus chemo.

### *Possible side effects*

Side effects of **ATRA** can include fever, headache, dry skin, mouth sores, skin rashes, swollen feet, itching and irritated eyes. It can also raise blood lipid levels (like cholesterol and triglycerides). Often blood liver test results become abnormal. These side effects often go away when the drug is stopped.

Most side effects of **ATO** are mild and can include feeling tired, nausea, vomiting, diarrhea, belly pain and nerve damage, called neuropathy, which can lead to numbness or tingling in the hands and feet. ATO can also cause problems with heart rhythm, which can sometimes be serious.

Both drugs can cause a serious side effect known as differentiation syndrome (previously called retinoic acid syndrome). This occurs when the leukemia cells release certain chemicals into the blood. It is most often seen during the first couple of weeks of treatment and in patients with a high white blood cell count. Symptoms can include fever, breathing problems, due to fluid buildup in the lungs and around the heart, low blood pressure, kidney damage, and severe fluid buildup elsewhere in the body. While differentiation syndrome can be serious, it can often be treated by stopping the drugs for a while and giving a steroid such as dexamethasone.

## FLT3 inhibitors

These drugs block proteins on leukemia cells that are important for cells to grow and divide, which causes leukemia cells to die. Examples of drugs that can block **FLT3** include **sorafenib (Nexavar), gilteritinib (Xospata)** and **midostaurin (Rydapt)**. These drugs can be used in the initial treatment of AML with *FLT3* gene changes or if the cancer comes back or does not respond well to treatment.

These drugs are taken by mouth in tablet or capsule form.

#### *Possible side effects*

Common side effects of these drugs include fever, mouth sores, feeling tired or weak, low blood counts, headaches or muscle aches, weight gain, stomach upset or pain, skin changes, and trouble sleeping.

Sorafenib can cause unique side effects. Some people treated with sorafenib develop high blood pressure, so doctors will often monitor the blood pressure during treatment. A distinct skin rash can occur with sorafenib as well. This causes redness, pain and sometimes swelling in the palms and soles of the feet.

### Menin inhibitors

In some children with AML, the leukemia cells have either a translocation in the *KMT2A* gene or a change (mutation) in the *NPM1* gene. Either of these gene changes can lead to the binding of the KMT2A protein with another protein called **menin**, which helps the cells grow.

**Revumenib (Revuforj)** is a menin inhibitor. It stops menin from combining with the KMT2A protein. This drug can be used to treat children with AML whose leukemia cells have a *KMT2A* translocation or an *NPM1* mutation, if the leukemia is no longer responding to treatment or has come back (relapsed) after other treatments. The child's blood or bone marrow can be tested to see if the leukemia cells have one of these gene changes.

This drug is taken by mouth as pills, typically twice a day.

#### *Possible side effects*

Common side effects of this drug can include nausea, vomiting, bleeding, diarrhea or constipation, fatigue, muscle pain, increased risk of infection, swelling in the arms and legs, and loss of appetite.

This drug may cause serious **heart rhythm problems**. This might lead to an irregular heartbeat, which could be life threatening. Your child's doctor will check blood mineral levels and get electrocardiograms (EKGs) to test your child's heart rhythm before and during treatment with this drug.

An important possible side effect of this drug is **differentiation syndrome**. This occurs when the leukemia cells release certain chemicals into the blood. It most often occurs shortly after starting treatment, but sometimes it can happen weeks or months later. Symptoms can include fever, coughing or breathing problems (from fluid buildup in the lungs and around the heart), dizziness or lightheadedness (from low blood pressure), urinating less often (from damage to the kidneys), and severe fluid buildup elsewhere in the body. It can often be treated by stopping the drug for a while and giving other medicines (such as dexamethasone or hydroxyurea).

## Targeted therapy drugs for chronic myeloid leukemia (CML)

Read about the targeted therapy drugs sometimes used to treat CML and their possible side effects.

## Tyrosine kinase inhibitors (TKIs)

Nearly all children with chronic myeloid leukemia (CML) have an abnormal chromosome in their leukemia cells known as the **Philadelphia chromosome**. These chromosomes have a specific gene change known as ***BCR-ABL***, which helps the leukemia cells multiply.

TKIs such as **imatinib (Gleevec), dasatinib (Sprycel), nilotinib (Tasigna),** and **bosutinib (Bosulif),** attack cells that have the *BCR-ABL* gene change. These drugs are very effective at controlling the leukemia for long periods of time in most children.

These drugs are taken daily as pills or capsules. Imatinib is available as a liquid, under the name Imkeldi.

### *Possible side effects*

Possible side effects of these drugs include diarrhea, nausea, muscle pain, fatigue, and skin rashes. These are generally mild. A common side effect is swelling around the eyes or in the hands or feet, which may be caused by the drugs' effects on the heart. Other possible side effects include lower blood counts when treatment starts. These drugs might also slow a child's growth, especially if used before puberty.

## More information about targeted therapy

To learn more about how targeted drugs are used to treat cancer, see Targeted Cancer Therapy[3].

To learn about some of the side effects listed here and how to manage them, see Managing Cancer-related Side Effects[4].

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects/infusion-immune-reactions.html
2. www.cancer.org/cancer/managing-cancer/side-effects/infusion-immune-reactions.html
3. www.cancer.org/cancer/managing-cancer/treatment-types/targeted-therapy.html
4. www.cancer.org/cancer/managing-cancer/side-effects.html

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# Immunotherapy for Childhood Leukemia

Immunotherapy is the use of medicines to help a person's own immune system recognize and destroy cancer cells. Many types of immunotherapies are being studied to treat childhood leukemia and some are now coming into use.

The types of immunotherapies used for treatment depend on the type of leukemia. Immunotherapy is not currently a part of treatment for children with forms of leukemia such as chronic myeloid leukemia (CML) or juvenile monomyelocytic leukemia (JMML).

Immunotherapy for acute lymphoblastic leukemia (ALL)
Immunotherapy for acute myeloid leukemia (AML)
More information about immunotherapy

## Immunotherapy for acute lymphoblastic leukemia (ALL)

Read about the immunotherapies sometimes used to treat ALL and their possible side effects.

### Chimeric antigen receptor (CAR) T-cell therapy

**Tisagenlecleucel (Kymriah)** is a CAR T-cell therapy that targets the CD19 protein on certain leukemia cells. It can be used to treat childhood ALL that has come back after treatment or that is no longer responding to treatment.

For this treatment, immune cells called **T cells** are removed from the child's blood through an IV line that goes into a machine during a process called **leukapheresis**. The remaining blood then goes back into the body. This typically takes a few hours, and it might need to be repeated.

The T cells are then genetically altered in the lab to have specific receptors (called **chimeric antigen receptors**, or CARs) on their surface. These receptors can attach to proteins on leukemia cells. The T cells are then multiplied in the lab and given back into the child's blood where they can seek out the leukemia cells and attack them. This process can take a few weeks.

For the treatment itself, the child typically gets chemo for a few days to help prepare the body. Then the CAR T cells are infused into a vein.

In most children who have had this treatment, the leukemia could no longer be detected within a few months of treatment, although it is not yet clear if this means that they have been cured. Many children still undergo a stem cell transplant after CAR T-cell therapy.

### *Possible side effects*

This treatment can have serious or even life-threatening side effects, which is why it needs to be given in a medical center that is specially trained in its use.

**Cytokine release syndrome (CRS):** CRS happens when T cells release chemicals (cytokines) that ramp up the immune system. This can happen within a few days to weeks after treatment and can be life-threatening. Symptoms can include:

- High fever and chills
- Trouble breathing
- Low blood pressure
- Heart problems
- Kidney problems
- Severe nausea, vomiting, and/or diarrhea
- Severe muscle or joint pain

- Feeling dizzy or lightheaded

**Nervous system problems:** This drug can have serious effects on the nervous system, which can result in symptoms such as:

- Headaches
- Changes in consciousness
- Confusion or agitation
- Seizures
- Trouble speaking and understanding
- Loss of balance

**Other serious side effects:** Other possible side effects can include:

- Serious infections
- Low blood cell count, which can increase the risk of infections, fatigue, and bruising or bleeding

It is very important to report any side effects to your child's health care team right away, as there are often medicines that can help treat them.

## Blinatumomab (Blincyto)

Blinatumomab is a special kind of monoclonal antibody known as a **bispecific T-cell engager (BiTE)**. It can attach to 2 different proteins at the same time. One part of blinatumomab attaches to the CD19 protein, which is found on B cells, including some leukemia cells. Another part attaches to CD3, a protein found on immune cells called T cells. By binding to both proteins, this drug brings the leukemia cells and immune cells together, which helps the immune system attack the leukemia.

This drug is used to treat some types of B-cell ALL in children. For example:

- It might be used as part of the second (consolidation) phase of treatment in some children with ALL.
- It might be used to treat ALL that comes back or that is no longer responding to other treatments.
- It might be used earlier during treatment for some infants with ALL.

Blinatumomab is given into a vein (IV) as a continuous infusion over 28 days. Because of certain serious side effects, such as cytokine release syndrome (see below) that may occur during the first few times it is given, the child usually needs to stay in the hospital for the first few days of the infusion.

***Possible side effects***

**Cytokine release syndrome (CRS):** CRS happens when the body releases chemicals (cytokines) that ramp up the immune system. Symptoms can include:

- High fever and chills
- Trouble breathing
- Low blood pressure
- Heart problems
- Kidney problems
- Severe nausea, vomiting, and/or diarrhea
- Severe muscle or joint pain
- Feeling dizzy or lightheaded

**Nervous system problems:** This drug can have serious effects on the nervous system, which can result in symptoms such as:

- Headaches
- Changes in consciousness
- Confusion or agitation
- Seizures
- Trouble speaking and understanding
- Loss of balance

Some children might have serious infusion reactions[1] (similar to an allergic reaction) while getting this drug. Symptoms can include feeling lightheaded or dizzy (due to low blood pressure), headache, nausea, fever or chills, shortness of breath, or wheezing. Your child will often be given medicines before the infusion starts to help prevent this.

## Inotuzumab ozogamicin (Besponsa)

Inotuzumab ozogamicin is an antibody-drug conjugate (ADC) used in some children with B cell ALL. This treatment is a monoclonal antibody attached to chemo drugs. The antibody part acts like a homing device, bringing the chemo drug directly to the leukemia cells.

This drug is described in more detail in Targeted Therapy Drugs for Childhood Leukemia.

## Rituximab (Rituxan)

Rituximab is a monoclonal antibody that targets a protein on many B cells, called CD20. When rituximab binds to leukemia cells with this protein, it triggers the cells to die. Some people with B cell ALL, especially Burkitt leukemia, may get rituximab as a part of their treatment plan.

Rituximab is given into a vein (IV).

#### Possible side effects

Because rituximab targets any cells with the CD20 protein, healthy B cells are also affected by the treatment. This can lead to a weaker immune system and lower levels of antibodies, sometimes for months to years after treatment. Other side effects of rituximab can include fevers, low energy, headaches, upset stomach, muscle or joint aches, flushing, and night sweats.

Some children might have serious infusion reactions[2] (similar to an allergic reaction) while getting this drug. Symptoms can include feeling lightheaded or dizzy (due to low blood pressure), headache, nausea, fever or chills, shortness of breath, and/or wheezing. Your child will be given medicines before each infusion to help prevent this.

A rare and severe blistering skin rash with mouth sores, called Stevens-Johnson syndrome, has also been seen after rituximab.

## Immunotherapy for acute myeloid leukemia (AML)

**Gemtuzumab ozogamicin (Mylotarg)**: This is an antibody-drug conjugate (ADC) used

in some children with AML. This treatment is a monoclonal antibody attached to chemo drugs. The antibody part acts like a homing device, bringing the chemo drug directly to the leukemia cells.

These medicines are described in more detail in Targeted Therapy Drugs for

Childhood Leukemia.

## More information about immunotherapy

To learn more about how drugs that work on the immune system are used to treat cancer, see Cancer Immunotherapy[3].

To learn about some of the side effects listed here and how to manage them, see Managing Cancer-related Side Effects[4].

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects/infusion-immune-reactions.html
2. www.cancer.org/cancer/managing-cancer/side-effects/infusion-immune-reactions.html
3. www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy.html
4. www.cancer.org/cancer/managing-cancer/side-effects.html

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Last Revised: July 22, 2025

# High-dose Chemotherapy and Stem Cell Transplant for Childhood Leukemia

A stem cell transplant (SCT), also known as a **bone marrow transplant** or **hematopoietic stem cell transplant**), can sometimes be used to help improve the chances of curing childhood leukemia.

How SCT is different from solid organ transplants
Allogeneic stem cell transplant
When a SCT might be used

Practical points
More information about stem cell transplant

## How SCT is different from solid organ transplants

Compared to solid organ transplants that work immediately, SCT instead “plants the seeds” that allow new bone marrow to grow and make blood cells. First, high-dose chemotherapy destroys the bone marrow (where new blood cells are formed) and the leukemia. Then, SCT replaces the blood-forming stem cells and immune system in the bone marrow that were killed by the chemo. It can take weeks to months after new blood stem cells are given before the bone marrow is able to make all of the blood cells on its own.

To learn about how a SCT is done, see Stem Cell Transplant for Cancer[1].

## Allogeneic stem cell transplant

For childhood leukemias, the type of transplant used is known as an allogeneic SCT. In this type of transplant, the blood-forming stem cells are donated from another person.

The donor’s tissue type (also known as the **HLA type**) should match the patient’s tissue type as closely as possible to help prevent the risk of major problems with the transplant. Tissue type is based on certain substances on the surface of cells in the body. The closer the tissue match between the donor and the recipient, the better the chance the transplanted cells will “take” and begin making new blood cells. This is different from a person’s blood type (also known as ABO or Rh).

The donor may be a sibling with the same tissue type as the patient. Other times, it can be an HLA-matched, unrelated donor—a stranger who has volunteered to donate blood-forming stem cells. Sometimes umbilical cord blood stem cells are used. These stem cells come from blood drained from the umbilical cord and placenta after a baby is born and the umbilical cord is cut. (This blood is rich in stem cells.) Whatever their source, the stem cells are then frozen and stored until they are needed for the transplant.

## When a SCT might be used

Whether a SCT may be used in treatment depends on the type of leukemia.

## Transplant in acute lymphoblastic leukemia (ALL)

In ALL, SCT might be used in children in some high-risk groups whose leukemia is more likely to come back after the initial (induction) chemo. In this case, the transplant is done after the induction chemo puts the leukemia into remission.

SCT might also be an option if the leukemia does not respond well to initial treatment, or if it relapses (comes back) soon after going into remission. It is less clear if SCT should be used for children whose ALL relapses later (such as more than 6 months or a year) after finishing the initial chemo. These children may respond well to another round of standard-dose chemo.

SCT may also be recommended for children with less common forms of ALL, such as those whose leukemias have the Philadelphia chromosome or those with T-cell ALL that don't respond well to initial treatment.

## Transplant in acute myelogenous leukemia (AML)

Because AML relapses more often than ALL, SCT might be recommended soon after the AML has gone into remission (after the initial chemo treatment), if the child has a donor who can donate stem cells for the transplant. This is especially true if there is a very high risk of relapse. This may be the case with some types of AML or when there are certain gene or chromosome changes in the leukemia cells. There is still some debate about which children with AML need this type of intensive treatment.

If a child with AML relapses after their first round of standard chemo, most doctors will recommend SCT as soon as the child goes into remission again.

In either case, it is important that the leukemia is in remission before getting a SCT. Otherwise, the leukemia is more likely to return.

## Transplant in chronic myeloid leukemia (CML)

For most children, SCT is not needed. Newer targeted therapy drugs work well for CML. Transplant is an option when usual treatments do not work well.

## Transplant in juvenile myelomonocytic leukemia (JMML)

JMML is rare and can be difficult to treat. SCT offers the best chance to cure JMML in most cases.

## Practical points

A SCT is a complex treatment that can cause life-threatening side effects[2]. If the doctors think your child can benefit from a transplant, the best place to have this done is at a cancer center where the staff has experience with the procedure and with managing the recovery period.

A SCT often requires a long hospital stay and can be very expensive. Even if the transplant is covered by your insurance, your co-pays or other costs could easily amount to many thousands of dollars. It is important to find out what your insurer will cover before the transplant to get an idea of what you might have to pay.

Be sure to talk to your child's doctor before the transplant to learn about possible long-term side effects your child might have. More information on long-term effects can be found in After Treatment for Childhood Leukemia.

## More information about stem cell transplant

To learn more about stem cell transplants, including how they are done and their potential side effects, see Stem Cell Transplant for Cancer[3].

For more general information about side effects and how to manage them, see Managing Cancer-related Side Effects[4].

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/stem-cell-transplant.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/stem-cell-transplant/transplant-side-effects.html

3. www.cancer.org/cancer/managing-cancer/treatment-types/stem-cell-transplant.html
4. www.cancer.org/cancer/managing-cancer/side-effects.html

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# Surgery for Childhood Leukemia

Surgery has a very limited role in treating childhood leukemia. Because leukemia cells spread widely throughout the bone marrow and blood, it is not possible to cure this type of cancer with surgery. Aside from a possible lymph node biopsy, surgery rarely has any role in diagnosing leukemia, since this is usually done with a bone marrow aspirate and biopsy.

Placing a central venous catheter
More information about surgery

## Placing a central venous catheter

Often before chemotherapy starts, a procedure is done to place a small plastic tube, called a central line (also known as a central venous catheter (CVC) or venous access[1] device (VAD)[2]), into a large blood vessel.

A central line is similar to an IV in the hand or arm, but central lines are bigger and can stay in the body for months. Central lines can make getting chemo and blood draws easier and less painful. It lowers the number of needlesticks needed during treatment. The health care team will teach you how to care for your child's central line to reduce the risk of problems, such as infection.

## More information about surgery

For more general information about surgery as a treatment for cancer, see Cancer Surgery[3].

To learn about some of the side effects listed here and how to manage them, see Managing Cancer-related Side Effects[4].

## Hyperlinks

1. www.cancer.org/cancer/supportive-care/iv-lines-ports.html

2. www.cancer.org/cancer/supportive-care/iv-lines-ports.html
3. www.cancer.org/cancer/managing-cancer/treatment-types/surgery.html
4. www.cancer.org/cancer/managing-cancer/side-effects.html

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# Radiation Therapy for Childhood Leukemia

Radiation therapy uses high-energy x-rays or particles to kill cancer cells.

When might radiation therapy be used?
How is radiation therapy given?
Possible side effects of radiation
More information about radiation therapy

## When might radiation therapy be used?

Today, radiation is not often needed to treat leukemia. However, it can be used in certain situations.

In any type of leukemia where a stem cell transplant is part of the treatment plan, radiation to the whole body, called total body irradiation (TBI), can be an important part of the treatment. (see High-Dose Chemotherapy and Stem Cell Transplant).

In some cases, it can be used to treat the spread of acute leukemia outside the bone marrow.

## Radiation therapy in acute lymphoblastic leukemia (ALL)

In ALL, cancer cells can spread to the brain or the testicles in males. When this occurs, radiation to the brain or testicle may be considered, especially if treatment with chemotherapy has not been successful. Radiation may be more likely to be used in situations where leukemia has come back (relapsed) after treatment in these areas.

## Radiation therapy in acute myeloid leukemia (AML)

In AML, cancer cells can also spread to the brain or collect outside the bone marrow in tumors called myeloid sarcomas or chloromas. When this occurs, radiation may be used to treat it, especially if treatment with chemotherapy has not been successful. Radiation may be more likely to be used in situations where leukemia has come back (relapsed) after treatment in these areas.

## How is radiation therapy given?

This type of treatment is given by a doctor called a **radiation oncologist**. Before treatments start, the radiation team will take careful measurements with imaging tests, such as a CT or MRI, during a planning session called a treatment **simulation**.

The treatment itself is like getting an x-ray, but the radiation is much stronger. Each treatment lasts only a few minutes. Most of the time for the treatment is spent making sure the radiation is aimed correctly. The treatment is painless, but some younger children may need to be given medicine beforehand to make sure they do not move during the treatment.

### Possible side effects of radiation

The side effects of radiation depend on where the radiation is given and the dose of radiation. The side effects during or shortly after treatment can include:

- Fatigue
- Hair loss in the treated area
- Skin that may be red, tender, or dry like a sunburn in the treated area
- Low blood counts
- Mouth sores, trouble swallowing, and/or loss of appetite (from treatment to the head and neck area)
- Nausea, vomiting, cramps, and/or diarrhea (from treatment to the belly)

Radiation can also have long-term side effects. These are described in After Treatment for Childhood Leukemia.

### More information about radiation therapy

To learn more about how radiation is used to treat cancer, see Radiation Therapy[1].

To learn about some of the side effects listed here and how to manage them, see Managing Cancer-related Side Effects[2].

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/radiation.html
2. www.cancer.org/cancer/managing-cancer/side-effects.html

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# Treatment of Children with Acute Lymphocytic Leukemia (ALL)

The main treatment for children with ALL is chemotherapy. Children with ALL are typically classified by risk group to make sure that the correct types and doses of drugs are given. Treatment depends on the risk group.

Phases of chemotherapy for ALL
Treating Philadelphia chromosome-type ALL
Treating Burkitt leukemia
Treating refractory or recurrent ALL

## Phases of chemotherapy for ALL

The entire length of treatment is typically about 2 to 3 years. Chemo for ALL is given in two main ways, systemically (through the blood stream or by mouth) and intrathecally (to the fluid that bathes the brain and spinal cord called the cerebrospinal fluid).

Chemotherapy for ALL is usually given in 3 main phases:

## Induction

The goal of induction chemotherapy is to achieve a **remission**. This means that leukemia cells are no longer found in bone marrow samples, the normal marrow cells return, and the blood counts improve. (A remission is not necessarily a cure.) More than 95% of children with ALL enter remission after 4 weeks of induction treatment.

This first month of treatment is intense and requires hospital stays and frequent visits to the doctor. Your child may spend some or much of this time in the hospital, because serious infections[1] or other complications can occur. It is very important to take all medicines as prescribed and tell your care team about any new symptoms, such as a fever. Sometimes complications can be serious enough to be life threatening, but in recent years, advances in supportive care (nursing care, nutrition, antibiotics, red blood cell and platelet transfusions as needed, etc.) have made these much less common than in the past.

Children with ALL may receive 3 or 4 chemo drugs for the first 4 weeks of treatment. These include asparaginase, vincristine, steroids (such as prednisone or dexamethasone) and, in some cases, a fourth drug in the anthracycline class (daunorubicin or doxorubicin). Whether 3 or 4 drugs are used may depend on the type of ALL, ALL risk group, and routine practice at your treatment center.

Children with Philadelphia chromosome-positive ALL may be given a targeted drug, such as imatinib or dasatinib, as well.

**Intrathecal chemotherapy:** All children with ALL also get chemo into the cerebrospinal fluid (CSF) to kill any leukemia cells that might have spread to the brain and spinal cord. This treatment, known as intrathecal chemotherapy, is given through a lumbar puncture (spinal tap). This treatment starts during induction and is often given in the later phases of treatment as well. How often intrathecal chemo is given depends on a few factors like risk group and whether or not leukemia cells were found in the CSF at the time the leukemia was diagnosed.

Some cancer centers use methotrexate or cytarabine for intrathecal chemo doses, and others may use a combination of methotrexate, hydrocortisone, and cytarabine (called triple intrathecal chemo). Triple intrathecal chemo is more common in the treatment of children with higher-risk ALL.

Along with intrathecal chemo, some high-risk patients (for example, those with T-cell ALL and those with many leukemia cells in their CSF when the leukemia is diagnosed) may be given radiation therapy to the brain. This was more common in the past, but recent studies have found that many children, even with high-risk ALL, may not need radiation therapy if they are given more intensive chemo. Doctors try to avoid giving radiation to the brain, especially in younger children, because no matter how low the dose is kept, it can cause problems with thinking, learning, growth, and development.

A possible side effect of intrathecal chemo is seizures during treatment, which happen in a small percentage of children. Children who develop seizures are treated with drugs to prevent them.

## Consolidation (intensification)

The consolidation phase of chemo starts once the leukemia is in remission and typically lasts for several months. This phase further reduces the number of leukemia cells still in the body. Several chemo drugs are combined to help prevent any remaining leukemia cells from developing resistance. Intrathecal chemo (as described above) is continued during this time.

During consolidation, children may be treated with drugs such as methotrexate, cyclophosphamide, cytarabine, 6-mercaptopurine (6-MP), 6-thioguanine (6-TG), vincristine, asparaginase, doxorubicin, and/or steroids, but regimens differ among cancer centers. Children with T-cell ALL may be treated with nelarabine during this phase as well.

Children with high-risk leukemia, because of gene or chromosome changes in the leukemia cells or because there is still minimal residual disease after induction, generally get more intense treatment and higher doses of certain chemo drugs over the treatment course.

For some children with ALL, there may be other rounds of intense chemotherapy as part of consolidation. These are known as **delayed intensification** and **interim maintenance**.

Children with Philadelphia chromosome-positive ALL may also get a targeted drug such as imatinib (Gleevec).

For some children with B-cell ALL, the immunotherapy drug blinatumomab (Blincyto) might be part of the consolidation phase as well.

For some children in high-risk groups, a stem cell transplant might be an option at this time once the leukemia is in remission.

## Maintenance

If the leukemia remains in remission after induction and consolidation, maintenance therapy can begin. Most treatment plans use daily 6-mercaptopurine (6-MP) and weekly methotrexate, given by mouth, often along with vincristine, which is given into a vein (IV), and a steroid (prednisone or dexamethasone). These latter 2 drugs are given for brief periods every 12 weeks. Other drugs may be added depending on the type of ALL and the risk of recurrence. Intrathecal chemo continues during maintenance treatment.

Maintenance can last up to 2 years. For most children, maintenance chemo can be given in the clinic or at home. Some children at higher risk may get more intense maintenance chemo and intrathecal therapy.

## Treating Philadelphia chromosome-type ALL

For children with certain types of ALL, such as those with the Philadelphia chromosome, standard chemotherapy for ALL (as outlined above) might not be as effective. A stem cell transplant may be advised if induction treatment puts the leukemia in remission and a suitable stem cell donor is available.

Newer, targeted drugs such as **imatinib (Gleevec)** and **dasatinib (Sprycel)** are designed to kill leukemia cells that have the Philadelphia chromosome. These drugs are taken as pills. Adding these drugs to chemotherapy throughout treatment seems to help improve outcomes, according to studies done so far.

## Treating Burkitt leukemia

Burkitt leukemia is a mature B-cell ALL, different from most other types of ALL in children. Burkitt leukemia is treated like Burkitt lymphoma[2], a type of non-Hodgkin lymphoma seen in children and young adults.

Treatment for Burkitt leukemia is shorter than treatment for other types of ALL, lasting several months in most cases. It includes treatment with chemo drugs such as cyclophosphamide, vincristine, prednisone, doxorubicin, cytarabine, methotrexate and etoposide. Treatment also includes the use of rituximab, which is an immunotherapy drug that targets B cells, and the use of intrathecal chemotherapy to treat and prevent leukemia in the fluid around the brain and spinal cord.

For more information, please see our page on the treatment of non-Hodgkin lymphoma for more information on the treatment of Burkitt lymphoma[3].

## Treating refractory or recurrent ALL

In some cases, ALL may not respond to the usual treatments. In this case, doctors may find leukemia on bone marrow or blood tests after treatment has started that does not go away. Sometimes, this is called **minimal residual disease (MRD)**, which means special lab tests must be used to find small amounts of leukemia. Other times, leukemia cells can be seen under the microscope on standard lab tests.

When leukemia cells never completely go away, this is called **refractory disease**. When the leukemia goes away during treatment and then recurs (comes back), this is called **recurrent disease** (or **relapse**).

Refractory disease may be treated with newer types of immunotherapy such as CAR T-cell therapy or blinatumomab (Blincyto), or the antibody-drug conjugate inotuzumab ozogamicin (Besponsa).

Recurrent disease during or after treatment is often treated again with chemotherapy. Much of the treatment depends on how soon the leukemia returns after the first treatment. If the relapse occurs after a long time, the same drugs might still be effective, so the same or similar treatment may be used to try to get the leukemia into a second remission.

If it comes back after a shorter time interval, more aggressive chemo with other drugs may be needed. The most used chemo drugs are vincristine, asparaginase, anthracyclines (doxorubicin, daunorubicin, or mitoxantrone), cyclophosphamide, cytarabine (ara-C), and either etoposide or teniposide. The child will also receive a steroid (prednisone or dexamethasone). Intrathecal chemo will also be given.

For children whose leukemia comes back soon after starting treatment, or for children with T-cell ALL who relapse, a stem cell transplant may be considered, especially if the child has a good donor. Stem cell transplants may also be used for children who relapse after a second course of chemotherapy.

Some children have an **extramedullary relapse**, meaning that leukemia cells are found in one part of the body, such as the cerebrospinal fluid (CSF) or the testicles, but are not detectable in the bone marrow. In addition to intensive chemotherapy as described above, children with spread to the CSF may get more intense intrathecal chemotherapy, sometimes with radiation to the brain and spinal cord (if that area had not been already treated with radiation). Boys with relapse in a testicle may get radiation to the area.

If ALL doesn't go away completely or if it comes back after a stem cell transplant, newer types of immunotherapy, such as CAR T-cell therapy or blinatumomab (Blincyto) or the antibody-drug conjugate inotuzumab ozogamicin (Besponsa) might still be options.

If the leukemia cells have a *KMT2A* gene change, treatment with a targeted drug called a menin inhibitor, such as revumenib (Revuforj), might be an option.

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects/infections.html
2. www.cancer.org/cancer/types/childhood-non-hodgkin-lymphoma/about/types-non-hodgkin-children.html
3. www.cancer.org/cancer/types/childhood-non-hodgkin-lymphoma/treating/by-stage.html

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# Treatment of Children with Acute Myeloid Leukemia (AML)

The main treatment for children with AML is chemotherapy. Because of the intensity of treatment and the risk of serious complications, children with AML need to be treated in cancer centers[1] or hospitals that have experience with this disease.

Phases of chemotherapy for AML
Treating refractory or recurrent AML
Treating children with acute promyelocytic leukemia (APL)

## Phases of chemotherapy for AML

Treatment of most children with acute myeloid leukemia (AML) is divided into 2 main phases of chemotherapy:

## Induction

The goal of induction chemotherapy is to achieve a **remission**. This means that leukemia cells are no longer found in bone marrow samples, the normal marrow cells return, and the blood counts improve. A remission is not necessarily a cure. More than 85% of children with AML enter remission after induction.

The chemo drugs most often used to treat AML early are cytarabine (ara-C), and anthracycline drugs (daunorubicin is used most often). Chemo treatments for AML are intense and cause very low blood counts. Children with AML often stay in the hospital during chemo and after, while the blood counts are low. Sometimes side effects from treatment can be serious enough to be life-threatening, but in recent years, advances in supportive care (nursing care, nutrition, antibiotics, red blood cell and platelet transfusions as needed, etc.) have made these much less common than in the past.

Some children with AML may get a dose of the targeted drug gemtuzumab ozogamicin (Mylotarg) along with chemo as part of their induction treatment. AML with *FLT3* gene changes may get targeted therapy as well with drugs like sorafenib (Nexavar).

If the doctors think that the leukemia may not respond well to just 2 chemo drugs alone, they may add another chemo drug such as etoposide or 6-thioguanine.

Treatment with these chemo drugs is repeated until the bone marrow shows no more leukemia cells. This usually occurs after 2 or 3 cycles of treatment.

**Preventing relapse in the central nervous system:** Most children with AML will also get intrathecal chemotherapy (given directly into the cerebrospinal fluid, or CSF) to treat or help prevent leukemia from relapsing in the brain or spinal cord. Radiation therapy to the brain is used less often.

## Consolidation (intensification)

Consolidation (intensification) begins after the induction phase. The purpose is to kill any remaining leukemia cells by using more intensive treatment.

Some children have a sibling who would be a good stem cell donor. For these children, a stem cell transplant might be recommended once the leukemia is in remission, especially if the AML has some poorer prognostic factors. Most studies have found this improves the chance for long-term survival over chemo alone, but it is also more likely to cause serious complications. For children with good prognostic factors, some doctors may recommend more cycles of intensive chemotherapy, and reserve the stem cell transplant in case the AML relapses.

Chemotherapy drugs used in this treatment phase may include etoposide, cytarabine, or anthracycline drugs (such as daunorubicin or mitoxantrone).

Intrathecal chemo (into the CSF) often continues during this phase of treatment.

Unlike treatment for ALL, maintenance chemo is not needed for most children with AML.

## Treating refractory or recurrent AML

In some cases, AML may not respond to the usual treatments. In this case, doctors may find leukemia on bone marrow or blood tests after treatment has started that does not go away. Sometimes, this is called **minimal residual disease (MRD)**, which means special lab tests must be used to find small amounts of leukemia. Other times, leukemia cells can be seen under the microscope on standard lab tests.

When leukemia cells never completely go away, this is called **refractory AML**. When the leukemia goes away during treatment and then recurs (comes back), this is called **recurrent (or relapsed) AML**.

Less than 10% of children have refractory AML (leukemia that does not respond to initial treatment). These leukemias are often very hard to cure, and doctors may recommend a stem cell transplant, if it can be done.

Generally, the outlook for a child whose AML relapses (comes back) after treatment is slightly better than if the AML never went into remission, but this depends on how long the initial remission was. In more than half of cases, relapsed leukemia can be put into a second remission with more chemo. The chance of getting a second remission is better if the first remission lasted for at least a year, but long-term second remissions are rare without a stem cell transplant. Many different combinations of standard chemo drugs have been used in these situations, but the results have been mixed.

Another option for some children with refractory or recurrent AML is treatment with targeted drugs such as gemtuzumab ozogamicin (Mylotarg) or venetoclax (Venclexta). If the leukemia cells have a *KMT2A* or *NPM1* gene change, treatment with a targeted drug called a menin inhibitor, such as revumenib (Revuforj), might be an option.

Most children whose leukemia has relapsed are good candidates for clinical trials[2] testing new treatment regimens. The hope is that remission can be achieved so that a stem cell transplant can be considered. Some doctors may advise a stem cell transplant even if there is no remission. This can sometimes be successful.

## Treating children with acute promyelocytic leukemia (APL)

Treatments used for acute promyelocytic leukemia (APL), a subtype of acute myeloid leukemia (AML), differs from the usual AML treatment. Similar to other forms of AML, there are two main phases of treatment: induction and consolidation.

During **induction**, children with APL typically respond very well to targeted drugs called arsenic trioxide (ATO) and all-trans retinoic acid (ATRA). This phase lasts at least 4 weeks, sometimes longer if the cancer cells are still detected in the blood. In children with white blood cell counts of less than 10,000 at diagnosis, called standard-risk APL, treatment with these drugs alone often results in remission. In high-risk APL, these drugs are used in combination with chemo (usually anthracycline drugs like daunorubicin or cytarabine).

Rarely, APL can spread to the cerebrospinal fluid (CSF). Lumbar punctures, when needed, are often delayed until any bleeding risk in the child is improved. Children with leukemia cells in the CSF will receive treatment with triple intrathecal therapy during this early phase of treatment. This treatment is a combination of cytarabine, hydrocortisone and methotrexate given into the CSF by a lumbar puncture. These treatments continue twice weekly for 2 to 3 weeks.

**Consolidation** is similar to induction, where ATO and ATRA are the main drugs used. The schedule of how these drugs are given is a little different than in the first phase of treatment. Because of how well this treatment works, stem cell transplant is not typically needed.

### Treating recurrent APL

If the leukemia comes back after treatment, it can be treated. ATO, ATRA, and gemtuzumab may be options for treatment. A stem cell transplant may be considered once a second remission is achieved.

## Hyperlinks

1. www.cancer.org/cancer/childhood-cancer/how-to-find-childhood-cancer-treatment.html
2. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/clinical-trials.html

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# Treatment of Children with Chronic Myeloid Leukemia (CML)

Chronic myeloid leukemia (CML) is rare in children, but it can occur. Treatment in children is like what is used for adults.

Common treatment approaches in childhood CML

## Common treatment approaches in childhood CML

Targeted drugs called tyrosine kinase inhibitors (TKIs), such as **imatinib (Gleevec), dasatinib (Sprycel), nilotinib (Tasigna)**, and **bosutinib (Bosulif)** attack cells with the Philadelphia chromosome, which is the key gene abnormality in CML cells. These drugs are usually good at controlling CML, often for long periods of time and with less severe side effects than chemo drugs. However, it is not yet clear if these drugs can cure CML when used alone, and they must be taken every day for a long time. Clinical trials are being done to help doctors decide if there is a safe timepoint to stop these drugs.

If targeted drugs are no longer helpful, high-dose chemotherapy with a stem cell transplant offers the best chance for a cure.

For more information on CML and its treatment, see Chronic Myeloid Leukemia.[1]

## Hyperlinks

1. www.cancer.org/cancer/types/chronic-myeloid-leukemia.html

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# Treatment of Children with Juvenile Myelomonocytic Leukemia (JMML)

Juvenile myelomonocytic leukemia (JMML) is rare so it has been hard to study which treatment might be best. There is no clear single best chemotherapy treatment for this leukemia.

Common treatment approaches for JMML

## Common treatment approaches for JMML

A stem cell transplant is the treatment of choice when possible, as it offers the best chance to cure JMML. Sometimes, even if the leukemia recurs, a second stem cell transplant can be helpful.

Because JMML is hard to treat with current chemo drugs, taking part in a clinical trial[1] looking at newer drugs may be a good option for children who cannot get a stem cell transplant.

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/clinical-trials.html

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# After Treatment for Childhood Leukemia

During treatment for childhood leukemia, the main concerns for most families are the daily aspects of getting through treatment and beating the leukemia. After treatment, the concerns may shift toward the long-term effects of the leukemia and its treatment, and concerns about the leukemia coming back.

It is normal to want to put the leukemia and its treatment behind you and to get back to a life that does not revolve around cancer. Getting the right follow-up care offers your child the best chance for recovery and long-term survival.

Follow-up exams and tests
Ask the treatment team for a survivorship care plan
Keeping health insurance and copies of medical records
Late and long-term effects of treatment
Social and emotional health after cancer
Long-term follow-up guidelines

## Follow-up exams and tests

For several years after treatment, regular follow-up exams will be very important. The doctors will watch for possible signs of leukemia coming back, as well as for short-term and long-term side effects of treatment[1].

Doctor visits often include careful physical exams and lab tests, and sometimes might include imaging tests. How often you follow up with your cancer care team will depend on the type of leukemia, treatment given, and other factors. Visits will usually be monthly during the first year, and then less often for at least 5 years after therapy. After that time, most children see their doctor at least yearly.

For the most common types of leukemia in children (ALL and AML), if the leukemia does come back, it is most often while the child is still being treated or within a year or so after finishing treatment. It is less common for ALL or AML to return if there are no signs of the disease within the next 2 years.

A benefit of follow-up care is that it gives you a chance to discuss questions and concerns that come up during and after your child's recovery. For example, almost any cancer treatment can have side effects. Some might last only a short time, but others can last longer or might not show up until months or years later. It is important to report any new symptoms to the doctor right away, so that the cause can be found and treated, if needed.

## Ask the treatment team for a survivorship care plan

Talk with the treatment team about developing a survivorship care plan[2]. This plan might include:

- A summary of the diagnosis, tests done, and treatment given
- A suggested schedule for follow-up exams and tests
- A schedule for other tests your child might need in the future, such as an ultrasound to monitor the heart (called an echocardiogram), or tests to look for other long-term health effects from the leukemia or its treatment
- A list of possible late or long-term side effects from your child's treatment, including what to watch for and when to contact the doctor
- Diet and physical activity suggestions

## Keeping health insurance and copies of medical records

As much as you might want to put the experience behind you once treatment is done, it is also important to keep good records of your child's medical care during this time. Eventually, your child will grow up, be on their own, and have new doctors. It is important for your child to be able to give the new doctors the details of their cancer diagnosis and treatment. Gathering the details soon after treatment may be easier than trying to get them at some point in the future.

Ask your child's cancer care team where and how to get this information. Learn more in Keeping Copies of Important Medical Records[3].

It is also very important to keep health insurance[4] coverage. Tests and doctor visits can cost a lot, and even though no one wants to think of the cancer coming back, this could happen.

## Late and long-term effects of treatment

Because of major progress in treating leukemia, most children will live long lives. Watching for late effects of treatment as children age is a vital part of their long-term health care.

Just as the treatment of childhood leukemia requires an individual approach, so does follow-up care after treatment. The earlier problems are recognized, the more likely they can be treated.

Many childhood leukemia survivors are at risk for several possible late effects of their treatment. This risk depends on a few factors, such as the type of leukemia and the type and doses of treatments they received. It is important to discuss what these health problems might be with your child's medical team, so you know what to watch for.

## Second cancers

Children who have been treated for leukemia are often at higher risk of developing other cancers later in life. One of the most serious possible side effects of ALL therapy is a small risk of getting AML later.

This occurs in a small percentage of patients after getting certain chemotherapy drugs, such as **epipodophyllotoxins** (etoposide, teniposide), **alkylating agents** (cyclophosphamide), or **anthracyclines** (daunorubicin, doxorubicin). Of course, the risk of getting these second cancers must be balanced against the benefit of treating a life-threatening disease such as leukemia.

## Heart problems

**Anthracycline** (daunorubicin, doxorubicin) and anthraquinone (mitoxantrone) chemo drugs can cause heart problems later in life. For some survivors, echocardiograms to look at the strength of the heart muscle may be recommended to catch and treat problems early.

## Learning problems

Treatment that includes **radiation therapy to the brain** or some types of chemotherapy (such as intrathecal chemotherapy, methotrexate or cytarabine) may affect learning ability in some children. Because of this, doctors try to limit treatments that could affect the brain as much as possible. Children treated for leukemia may need extra support in school.

## Growth and development

Some cancer treatments may affect a child's growth, so they might end up a bit shorter as adults. This is especially true after **radiation** or **stem cell transplants**. **Tyrosine kinase inhibitors** (TKIs) used for CML can also cause slowed growth. In some cases, this can be treated with growth hormone. Treatment might also affect the levels of other hormones in the body, which can increase the risk of health issues such as thyroid problems, obesity, and diabetes.

## Fertility issues

Alkylating chemo drugs (**cyclophosphamide**) may also affect sexual development and the ability to have children later in life. Talk to your child's cancer care team about the risks of infertility with treatment and ask if there are options for preserving fertility, such as sperm banking or egg freezing. For more information, see Preserving Fertility in Children and Teens With Cancer[5].

## Bone problems

Bone damage or osteoporosis (thinning of the bones) may result from the use of **steroids**, like prednisone and dexamethasone, during treatment.

## Nerve problems

Some children can develop neuropathy, or damage to the nerves that control how our body moves and feels, after treatment with drugs like **vincristine**. Symptoms of neuropathy, like numbness, tingling, or tripping while walking often go away or get better once treatment is done. For some children, these symptoms may be lifelong.

## Social and emotional health after cancer

Once treatment is finished, a number of emotional concerns can come up. Some of these might last a long time and can include:

- Dealing with physical changes that can result from the treatment
- Worries about the leukemia returning or new health problems developing
- Feelings of resentment for having had leukemia or having had to go through treatment when others did not
- Feeling guilty for surviving cancer, when other friends with cancer did not
- Concerns about being treated differently or discriminated against (by friends, classmates, coworkers, employers, etc.)
- Concerns about dating, marrying, and having a family later in life

It is normal to have some anxiety or other emotional reactions after treatment, but feeling overly worried, depressed, or angry can affect many aspects of a young person's growth. It can get in the way of relationships, school, work, and other aspects of life. With support from family, friends, other survivors, mental health professionals, and others, many people who have survived leukemia can thrive despite the challenges they have had to face.

There may be other possible complications from chemotherapy or other treatments as well. Your child's doctor should carefully review any possible problems with you before your child starts treatment.

## Long-term follow-up guidelines

To help increase awareness of late effects and improve follow-up care for childhood cancer survivors throughout their lives, the Children's Oncology Group (COG) has developed long-term follow-up guidelines for survivors of childhood cancers. These guidelines are used by your doctor to create a survivorship care plan specific to your child and their treatment. They can help you know what to watch for, what types of screening tests should be done, and how late effects can be treated.

To learn more, ask your child's doctors about the COG survivor guidelines. You can also read them on the COG website: www.survivorshipguidelines.org[6]. The guidelines are written for health care professionals, but patient education on some late effects of treatment is available (as "Health Links") on the site as well.

For more on the possible long-term effects of treatment, see Late and Long-term Effects of Childhood Cancer Treatment[7].

## Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects.html
2. www.cancer.org/cancer/survivorship/long-term-health-concerns/importance-of-follow-up-care/survivorship-care-plans.html
3. www.cancer.org/cancer/survivorship/long-term-health-concerns/keeping-copies-of-important-medical-records.html

4. www.cancer.org/cancer/financial-insurance-matters/understanding-health-insurance.html
5. www.cancer.org/cancer/managing-cancer/side-effects/fertility/preserving-fertility-in-children-and-teens-with-cancer.html
6. www.survivorshipguidelines.org/
7. www.cancer.org/cancer/childhood-cancer/late-effects-of-childhood-cancer-treatment.html

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